

EXERCISES

LEVEL I

(Problems based on Fundamentals)

ABC OF COORDINATES

- Find the polar co-ordinates of the points whose cartesian co-ordinates are $(3, -4)$ and $(-3, 4)$.
- Transform the equation $r^2 = a^2$ into cartesian form.
- Transform the equation $r = 2a \cos \theta$ into cartesian form.
- Transform the equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ into polar form.
- Transform the equation $2x^2 + 3xy + 2y^2 = 1$ into polar form.

DISTANCE BETWEEN TWO POINTS

- Find the distance between the points $(a \cos \alpha, a \sin \alpha)$ and $(a \cos \beta, a \sin \beta)$, where $a > 0$.
- Find the distance between the points $\left(3, \frac{\pi}{4}\right)$ and $\left(7, \frac{5\pi}{4}\right)$.
- If the point (x, y) be equidistant from the points $(6, -1)$ and $(2, 3)$ such that $Ax + By + C = 0$, find the value of $A + B + C + 10$.
- The vertices of a triangle ABC are $A(-2, 3)$, $B(2, -1)$ and $C(4, 0)$. Find $\cos A$.
- Prove that the points $(-4, -1)$, $(-2, -4)$, $(4, 0)$ and $(2, 3)$ are the vertices of a rectangle.
- Two vertices of an equilateral triangle are $(3, 4)$ and $(-2, 3)$. Find the co-ordinates of the third vertex.

SECTION FORMULAE

- Find the point, which divides the line joining the points $(2, 3)$ and $(5, -3)$ in the ratio $1 : 2$.
- In what ratio does y -axis divide the line segment joining $A(-3, 5)$ and $B(7, 2)$.
- Find the ratio in which the join of the points $(1, 2)$ and $(-2, 3)$ is divided by the line $3x + 4y = 7$.
- Find the co-ordinates of the points which trisect the line segment joining $(1, -2)$ and $(-3, 4)$.
- The co-ordinates of the mid-points of the sides of a triangle are $(1, 1)$, $(3, 2)$ and $(4, 1)$. Find the co-ordinates of its vertices.
- The co-ordinates of three consecutive vertices of a parallelogram are $(1, 3)$, $(-1, 2)$ and $(2, 5)$. Find its fourth vertex.
- Find the centroid of $\triangle ABC$, whose vertices are $A(2, 4)$, $B(6, 4)$, $C(2, 0)$.
- Two vertices of a triangle are $(-1, 4)$ and $(5, 2)$. If its centroid is $(0, -3)$, find its third vertex.
- Find the in-centre of $\triangle ABC$, whose vertices are $A(1, 2)$, $B(2, 3)$ and $C(3, 4)$.
- If a triangle has its orthocentre at $(1, 1)$ and circum-centre at $(3/2, 3/4)$, find its centroid.

- The vertices of a $\triangle ABC$ are $A(0, 0)$, $B(0, 2)$ and $C(2, 0)$. Find the distance between the circum-centre and orthocentre.

AREA OF A TRIANGLE

- Find the area of a triangle whose vertices are $(3, -4)$, $(7, 5)$ and $(-1, 10)$.
- Find the area of a triangle whose vertices are $(t, t+2)$, $(t+3, t)$ and $(t+2, t+2)$.
- If $A(x, y)$, $B(1, 2)$ and $C(2, 1)$ be the vertices of a triangle of area 6 sq.u., prove that $x + y + 9 = 0$.
- Find the area of a quadrilateral whose vertices are $(1, 1)$, $(7, -3)$, $(12, 2)$ and $(7, 21)$.
- Find the area of a pentagon whose vertices are $(4, 3)$, $(-5, 6)$, $(0, 0)$, $(3, -6)$ and $(-7, -2)$.
- Prove that the points $(a, b+c)$, $(b, c+a)$ and $(c, a+b)$ are collinear.
- Let the co-ordinates of A, B, C and D are $(6, 3)$, $(-3, 5)$, $(4, -2)$ and $(x, 3x)$, respectively and $\frac{\Delta DBC}{\Delta ABC} = \frac{1}{2}$, find x .
- Find the area of a triangle formed by the lines $7x - 2y + 10 = 0$, $7x + 2y - 10 = 0$ and $9x + y + 2 = 0$.

LOCUS OF A POINT

- If the co-ordinates of a variable point P be $(a \cos \theta, a \sin \theta)$, where θ is a parameter, find the locus of P .
- If the co-ordinates of a variable point P be $(at^2, 2at)$, find the locus of P .
- Find the locus of a movable point P , where its distance from the origin is 3 times its distance from y -axis.
- If a point moves in a plane in such a way that its distance from the point $(a, 0)$ to its distance is equal to its distance from y -axis.
- If the co-ordinates of a variable point P be $\left(t + \frac{1}{t}, t - \frac{1}{t}\right)$, where t is a parameter, find the locus of P .
- Find the locus of a movable point P , for which the sum of its distance from $(0, 3)$ and $(0, -3)$ is 8.
- A stick of length l slides with its ends on two mutually perpendicular lines. Find the locus of the mid-point of the stick.
- If O be the origin and A be a point on the line $y^2 = 8x$. Find the locus of the mid-point of OA .
- If P be the mid-point of the straight line joining the points $A(1, 2)$ and Q where Q is a variable point on the curve $x^2 + y^2 + x + y = 0$. Find the locus of P .
- A circle has the centre $(2, 2)$ and always touches x and y axes. If it always touches a line AB (where A and B lie on positive x and y axes), find the locus of the circum-centre of the $\triangle OAB$, where O is the origin.
- Find the locus of a movable point P , for which the difference of its distances from $(2, 0)$ and $(-2, 0)$ is 6.

42. If $A(1, 1)$ and $B(-2, 3)$ are two fixed points, find the locus of a point P so that the area of $\triangle PAB$ is 9 sq.u.
43. Find the locus of a point whose co-ordinates are given by $x = t^2 + t$, $y = 2t + 1$, where t is parameter.
44. If a variable line $\frac{x}{a} + \frac{y}{b} = 1$ intersects the axes at A and B , respectively such that $a^2 + b^2 = 4$ and O be the origin, find the locus of the circumcentre of $\triangle OAB$.
45. A variable line cuts the x and y axes at A and B respectively where $OA = a$, $OB = b$ (O the origin) such that $a^2 + b^2 = 27$, find the locus of the centroid of $\triangle OAB$.

TRANSFORMATION OF COORDINATES

46. Find the equation of the curve $2x^2 + y^2 - 3x + 5y - 8 = 0$, when the origin is shifted to the point $(-1, 2)$ without changing the direction of the axes.
47. The equation of a curve referred to the new axes retaining their directions and origin is $(4, 5)$ is $x^2 + y^2 = 36$. Find the equation referred to the original axes.
48. At what point, the origin be shifted if the co-ordinates of a point $(-1, 8)$ become $(-7, 3)$?
49. If the axes are turned through 45° , find the transformed form of the equation $3x^2 + 3y^2 + 2xy = 2$.
50. Transform to parallel axes through the point $(1, -2)$, the equation $y^2 - 4x + 4y + 8 = 0$.
51. If a point $P(1, 2)$ is translated itself 2 units along the positive direction of x -axis and then it is rotated about the origin in anti-clockwise sense through an angle of 90° , find the new position of P .
52. If the axes are rotated through $\frac{\pi}{4}$, the equation $x^2 + y^2 = a^2$ is transformed to $\lambda xy + a^2 = 0$, find the value of $\lambda = 10$.
53. Transform to axes inclined at $\frac{\pi}{3}$ to the original axes, the equation $x^2 + 2\sqrt{3}y - y^2 = 2a^2$.
54. What does the equation $2x^2 + 2xy + 3y^2 - 18x - 22y + 50 = 0$ become when referred to new rectangular axes through the point $(2, 3)$, the new set making $\frac{\pi}{4}$ with the old?
55. If a point $P(2, 3)$ is rotated through an angle of 90° about the origin in anti-clockwise sense, say at Q , find the co-ordinates of Q .
56. If a point $P(3, 4)$ is rotated through an angle of 60° about the point $Q(2, 0)$ in anti-clockwise sense, say at R , find the co-ordinates of R .

STRAIGHT LINE

LEVEL I

(Problems Based on Fundamentals)

ABC OF STRAIGHT LINE

1. Find the slope of the line PQ , where $P(2, 4)$ and $Q(3, 10)$.
2. Find the value of λ , if 2 is slope of the line through $(2, 5)$ and $(\lambda, 7)$.

3. Prove that the line joining the points $(2, -3)$ and $(-5, 1)$ is parallel to the line joining $(7, -1)$ and $(0, 3)$.
4. Prove that the points $(a, b + c)$, $(b, c + a)$ and $(c, a + b)$ are collinear.
5. Find the angle between the lines joining the points $(0, 0)$, $(2, 2)$ and $(2, -2)$ $(3, 5)$.
6. If the angle between two lines is 45° and the slope of one of them is $1/2$, find the slope of the other line.
7. Find the equation of a line passing through the point $(2, -3)$ and is parallel to x -axis.
8. Find the equation of a line passing through the point $(3, 4)$ and is perpendicular to y -axis.
9. Find the equation of a line which is equidistant from the lines $x = 6$ and $x = 10$.

VARIOUS FORMS OF A STRAIGHT LINE

10. Find the equation of a straight line which cuts off an intercept of 7 units on y -axis and has the slope 3.
11. Find the equation of a straight line which makes an angle of 135° with the positive direction of x -axis and cuts an intercept of 5 units on the positive direction of y -axis.
12. Find the equation of a straight line which makes an angle of $\tan^{-1}\left(\frac{3}{5}\right)$ with the positive direction of x -axis and cuts an intercept of 6 units in the negative direction of y -axis.
13. Find the equation of a straight line which cuts off an intercept of 4 units from y -axis and are equally inclined with the axes.
14. Find the equations of bisectors of the angle between the co-ordinate axes.
15. Find the equation of a line passing through the point $(2, 3)$ and making an angle of 120° with the positive direction of x -axis.
16. Find the equation of the right bisectors of the line joining the points $(1, 2)$ and $(5, 7)$.
17. Find the equation of a line which passes through the point $(1, 2)$ and makes an angle θ with the positive direction of x -axis, where $\cos \theta = -\frac{3}{5}$.
18. A line passes through the point $A(2, 0)$ which makes an angle of 30° with the positive direction of x -axis and is rotated about A in clockwise direction through an angle of 15° . Find the equation of the straight line in the new position.
19. Find the equation of a line passing through the points $(1, 2)$ and $(3, 4)$.
20. The vertices of a triangle are $A(10, 4)$, $B(-4, 9)$ and $C(-2, -1)$. Find the equation of its altitude through A .
21. The vertices of a triangle are $A(1, 2)$, $B(2, 3)$ and $C(5, 4)$. Find the equation of its median through A .
22. Find the equation of the internal bisector of $\angle BAC$ of the $\triangle ABC$, whose vertices are $A(5, 2)$, $B(2, 3)$ and $C(6, 5)$.

23. A square is inscribed in a $\triangle ABC$, whose co-ordinates are $A(0, 0)$, $B(2, 1)$ and $C(3, 0)$. If two of its vertices lie on the side AC , find the vertices of the square.
24. The line joining the points $A(2, 0)$ and $B(3, 1)$ is rotated about A in the anti-clockwise direction through an angle of 15° . Find the equation of a line in the new position.
25. Find the equation of a line which passes through the point $(3, 4)$ and makes equal intercepts on the axes.
26. Find the equation of a line which passes through the point $(2, 3)$ and whose x -intercept is twice of y -intercept.
27. Find the equation of a line passes through the point $(2, 3)$ so that the segment of the line intercepted between the axes is bisected at this point.
28. Find the equation of a line passing through the point $(3, -4)$ and cutting off intercepts equal but of opposite signs from the two axes.
29. Find the equation of a line passing through the point $(3, 2)$ and cuts off intercepts a and b on x - and y -axes such that $a - b = 2$.
30. Find the area of a triangle formed by lines $xy = 0$ and $2x + 3y = 6$.
31. Find the equation of a straight line passing through the point $(3, 4)$ so that the segment of the line intercepted between the axes is divided by the point in the ratio $2 : 3$.
32. Find the equation of the straight line which passes through the origin and trisect the intercept of the line $3x + 4y = 12$.
33. Find the area of a triangle formed by the lines $xy - x - y + 1 = 0$ and $3x + 4y = 12$.
34. A straight line cuts off intercepts from the axes of co-ordinates, the sum of the reciprocals of which is a constant. Show that it always passes through a fixed point.
35. The length of the perpendicular from the origin to a line is 5 and the line makes angle of 60° with the positive direction of y -axis. Find the equation of the line.
36. Find the equation of the straight line upon which the length of the perpendicular from the origin is 2 and the slope of the perpendicular is $5/12$.

DISTANCE FORM OF A STRAIGHT LINE

37. A line passing through the point $(3, 2)$ and making an angle θ with the positive direction of x -axis such that $\tan \theta = 3/4$. Find the co-ordinates of the point on the line that are 5 units away from the given point.
38. Find the co-ordinates of the points at a distance $4\sqrt{2}$ units from the point $(-2, 3)$ in the direction making an angle of 45° with the positive direction of x -axis.
39. A point $P(3, 4)$ moves in a plane in the direction of $\hat{i} + \hat{j}$. Find the new position of P .
40. A line joining two points $A(2, 0)$ and $B(3, 1)$ is rotated about A in anti-clockwise direction through an angle of 15° . Find the new position of B .
41. Find the direction in which a straight line must be drawn through the point $(1, 2)$ so that its point of inter-

section with the line $x + y = 4$ may be at a distance $\frac{\sqrt{2}}{3}$ from the point $(1, 2)$.

42. The centre of a square is at the origin and one vertex is $P(2, 1)$. Find the co-ordinates of other vertices of the square.
43. The extremities of a diagonal of a square are $(1, 1)$ and $(-2, -1)$. Find the other two vertices.
44. A line through $(2, 3)$ makes an angle $\frac{3\pi}{4}$ with the negative direction of x -axis. Find the length of the line segment cut off between $(2, 3)$ and the line $x + y = 7$.
45. If the straight line drawn through the point $P(\sqrt{3}, 2)$ and making an angle of $\frac{\pi}{6}$ with the x -axis meets the line $\sqrt{3}x - 4y + 8 = 0$ at Q . Find the length of PQ .
46. Find the distance of the point $(2, 3)$ from the line $2x - 3y + 9 = 0$ measured along the line $2x - 2y + 5 = 0$.
47. Find the distance of the point $(3, 5)$ from the line $2x + 3y = 14$ measured parallel to the line $x - 2y = 1$.
48. Find the distance of the point $(2, 5)$ from the line $3x + y + 4 = 0$ measured parallel to the line $3x - 4y + 8 = 0$.
49. A line is drawn through $A(4, -1)$ parallel to the line $3x - 4y + 1 = 0$. Find the co-ordinates of the two points on this line which are at a distance of 5 units from A .

REDUCTION OF A STRAIGHT LINE

50. Reduce $x + \sqrt{3}y + 4 = 0$ into the
 - (i) slope intercept form and also find its slope and y -intercept.
 - (ii) intercept form and also find the lengths of x and y intercepts.
 - (iii) normal form and also find the values of p and α .

PARALLEL/PERPENDICULAR FORM OF A STRAIGHT LINE

51. Find the location of the points $(2, 2)$ and $(3, 5)$ with respect to the line
 - (i) $2x + 3y + 4 = 0$
 - (ii) $3x - 2y + 2 = 0$
 - (iii) $x + y - 7 = 0$
52. Determine whether the point $(2, -7)$ lies on the origin side of the line $2x + y + 2 = 0$ or not.
53. Write the parallel line to each of the following lines.
 - (i) $3x - 4y + 10 = 0$
 - (ii) $2x + 5y + 10 = 0$
 - (iii) $-x + y - 2012 = 0$
 - (iv) $y = x$
 - (v) $x = 5$
 - (vi) $y = 2013$
54. Find the equation of a line parallel to $3x + 4y + 5 = 0$ and passes through $(2, 3)$.
55. Find the equation of a line parallel to $3x - 4y + 6 = 0$ and passing through the mid-point of the line joining the points $(2, 3)$ and $(4, -1)$.
56. Find the equation of a line passing through $(2, 1)$ and parallel to the line joining the points $(2, 3)$ and $(3, -1)$.
57. Write the perpendicular line to each of the following lines.
 - (i) $5x - 3y + 2010 = 0$
 - (ii) $3x + 5y + 2011 = 0$

- (iii) $-x + y - 2012 = 0$ (iv) $y = 2011x$
 (v) $x = 2014$ (vi) $y = 2013$
58. Find the equation of a line perpendicular to $2x + 3y - 2012 = 0$ and passing through $(3, 4)$.
59. Find the equation of a line perpendicular to $2x - 3y - 5 = 0$ and cutting an intercept 1 on the x -axis.
60. Find the equation of the right bisectors of the line joining the points $(1, 2)$ and $(3, 5)$.
61. Find the equation of the altitude through A of $\triangle ABC$, whose vertices are $A(1, 2)$, $B(4, 5)$ and $C(3, 8)$.
62. Find the equation of a line passing through the point of intersection of the lines $2x + y = 8$ and $x - y = 10$ and is perpendicular to $3x + 4y + 2012 = 0$.
63. Find the distance of the point $(4, 5)$ from the straight line $3x - 5y + 7 = 0$.
64. The equation of the base of an equilateral triangle be $x + y = 2$ and one vertex is $(2, -1)$. Find the length of the sides of the triangle.
65. If a and b be the intercepts of a straight line on the x and y axes, respectively and p be the length of the perpendicular from the origin, prove that $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2}$.
66. Find the distance between the lines $3x - 4y = 5$ and $6x - 8y + 11 = 0$.
67. Let L has intercepts a and b on the co-ordinate axes. When the axes are rotated through an angle, keeping the origin fixed, the same line L has intercepts p and q , prove that $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2} + \frac{1}{q^2}$.

AREA OF A PARALLELOGRAM

68. If the area of a parallelogram formed by the lines $x + 3y = a$, $3x - 2y + 3a = 0$, $x + 3y + 4a = 0$ and $3x - 2y + 7a = 0$ is ma^2 , find the value of $11m + 30$.
69. Prove that the four lines $ax \pm by \pm c = 0$ enclose a rhombus, whose area is $\frac{2c^2}{|ab|}$.

FAMILY OF STRAIGHT LINES

70. Find the point of intersection of the lines $x - y + 4 = 0$ and $2x + y = 10$.
71. Prove that the three lines $2x - 3y + 5 = 0$, $3x + 4y = 7$ and $9x - 5y + 8 = 0$ are concurrent.
72. Find the equation of a line which is passing through the point of intersection of the lines $x + 3y - 8 = 0$, $2x + 3y + 5 = 0$ and $(1, 2)$.
73. Find the value of m so that the lines $y = x + 1$, $2x + y = 16$ and $y = mx - 4$ may be concurrent.
74. If the lines $ax + y + 1 = 0$, $x + by + 1 = 0$ and $x + y + c = 0$ (where a, b, c are distinct and different from 1) are concurrent, find the value of $\frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c}$.
75. If $2a + 3b + 6c = 0$, the family of straight lines $ax + by + c = 0$ passes through a fixed point. Find the co-ordinates of the fixed point.

78. If $4a^2 + 9b^2 - c^2 + 12ab = 0$, the family of the straight lines $ax + by + c = 0$ is either concurrent at (m, n) or (p, q) . Find the value of $m + n + p + q + 10$.
79. The family of lines $x(a + 2b) + y(a - 3b) = a - b$ passes through a fixed point for all values of a and b . Find the co-ordinates of the fixed point.
80. Find the equation of a line passing through the point of intersection of $2x + 3y + 1 = 0$, $3x - 5y - 5 = 0$ and equally inclined to the axes.
81. Find the slope of the lines which make an angle of 45° with the line $3x - y + 5 = 0$.

EQUATION OF STRAIGHT LINES PASSING THROUGH A GIVEN POINT AND MAKING A GIVEN ANGLE WITH A GIVEN LINE

82. Find the equations of the lines through the point $(2, 3)$ which make an angle 45° with the line $x - 2y = 3$.
83. A vertex of an equilateral triangle is $(2, 3)$ and the equation of the opposite side is $x + y = 2$. Find the equation of the other sides of the triangle.
84. A line $4x + y = 1$ through the point $A(2, -7)$ meets the line BC , whose equation is $3x - 4y + 1 = 0$ at the point B . Find the equation of the line AC so that $AB = AC$.
85. Find the equations of straight lines passing through $(-2, -7)$ and having an intercept of length 3 between the straight lines $4x + 3y = 12$ and $4x + 3y = 3$.
86. Find the equations of the lines passing through the point $(2, 3)$ and equally inclined to the lines $3x - 4y = 7$ and $12x - 5y + 6 = 0$.
87. Two straight lines $3x + 4y = 5$ and $4x - 3y = 15$ intersect at A . Points B and C are chosen on these lines such that $AB = AC$. Determine the possible equations of the line BC passing through the point $(1, 2)$.
88. Two equal sides of an isosceles triangle have the equations $7x - y + 3 = 0$ and $x + y = 3$ and its third side passes through the point $(1, -10)$. Find the equation of the third side.

EQUATIONS OF BISECTORS

89. Find the equations of the bisectors of the angle between the straight lines $3x - 4y + 7 = 0$ and $12x + 5y - 2 = 0$.
90. Find the equation of the bisectors bisecting the angle containing the origin of the straight lines $4x + 3y = 6$ and $5x + 12y + 9 = 0$.
91. Find the bisector of the angle between the lines $2x + y = 6$ and $2x - 4y + 7 = 0$, which contains the point $(1, 2)$.
92. Find the equation of the bisector of the obtuse angle between the lines $3x - 4y + 7 = 0$ and $12x + 5y = 2$.
93. Find the bisector of the acute angle between the lines $x + y = 3$ and $7x - y + 5 = 0$.
94. Prove that the length of the perpendiculars drawn from any point of the line $7x - 9y + 10 = 0$ to the lines $3x + 4y = 5$ and $12x + 5y = 7$ are the same.
95. Find the co-ordinates of the incentre of the triangle whose sides are $x + 1 = 0$, $3x - 4y = 5$, $5x + 12y = 27$.
96. The bisectors of the angle between the lines $y = \sqrt{3}x + 3$ and $\sqrt{3}y = x + 3\sqrt{3}$ meet the x -axis respectively, at P and Q . Find the length of PQ .

97. Two opposite sides of a rhombus are $x + y = 1$ and $x + y = 5$. If one vertex is $(2, -1)$ and the angle at that vertex be 45° . Find the vertex opposite to the given vertex.
98. Find the foot of perpendicular drawn from the point $(2, 3)$ to the line $y = 3x + 4$.

IMAGE OF A POINT WITH RESPECT TO A STRAIGHT LINE

99. Find the image of the point $(-8, 12)$ with respect to the line mirror $4x + 7y + 13 = 0$.
100. Find the image of the point $(3, 4)$ with respect to the line $y = x$.
101. If $(-2, 6)$ be the image of the point $(4, 2)$ with respect to the line $L = 0$, find the equation of the line L .
102. The image of the point $(4, 1)$ with respect to the line $y = x$ is P . If the point P is translated about the line $x = 2$, the new position of P is Q . Find the co-ordinates of Q .
103. The image of the point $(3, 2)$ with respect to the line $x = 4$ is P . If P is rotated through an angle $\frac{\pi}{4}$ about the origin in anti-clockwise direction. Find the new position of P .
104. The equations of the perpendicular bisectors of the sides AB and AC of $\triangle ABC$ are $x - y + 5 = 0$ and $x + 2y = 0$, respectively. If the point A is $(1, -2)$, find the equation of the line BC .

REFLECTION OF A STRAIGHT LINE

105. A ray of light is sent along the line $x - 2y = 3$. Upon reaching the line $3x - 2y = 5$, the ray is reflected from it. Find the equation of the line containing the reflected ray.
106. A ray of light passing through the point $(1, 2)$ is reflected on the x -axis at a point P and passes through the point $(5, 3)$. Find the abscissa of the point P .
107. A ray of light is travelling along the line $x = 1$ and gets reflected from the line $x + y = 1$, find the equation of the line which the reflected ray travel.
108. A ray of light is sent along the line $x - 6y = 8$. After refracting across the line $x + y = 1$, it enters the opposite side after turning by 15° away from the line $x + y = 1$. Find the equation of the line along with the reflected ray travels.

Hints & Solutions

Level - 1

(RECTANGULAR CARTESIAN CO-ORDINATES)

1. We have,

$$(3, -4) \Rightarrow \left(\sqrt{3^2 + (-4)^2}, \tan^{-1}\left(-\frac{4}{3}\right) \right)$$

$$\Rightarrow \left(5, -\tan^{-1}\left(\frac{4}{3}\right) \right)$$

and

$$(-3, 4) \Rightarrow \left(\sqrt{(-3)^2 + (4)^2}, \tan^{-1}\left(-\frac{3}{4}\right) \right)$$

$$\Rightarrow \left(5, \pi - \tan^{-1}\left(\frac{4}{3}\right) \right)$$

2. We have,

$$r^2 = a^2 \cos 2\theta$$

$$\Rightarrow r^2 = a^2 (\cos^2 \theta - \sin^2 \theta)$$

$$\Rightarrow r^2 = a^2 \left(\frac{x^2}{r^2} - \frac{y^2}{r^2} \right)$$

$$\Rightarrow \begin{aligned} r^2 &= a^2(x^2 - y^2) \\ (x^2 + y^2)^2 &= a^2(x^2 - y^2) \end{aligned}$$

3. We have,

$$r = 2a \cos 2\theta$$

$$\Rightarrow r = 2a \left(\frac{x}{r} \right)$$

$$\Rightarrow r^2 = 2ax$$

$$\Rightarrow (x^2 + y^2) = 2ax$$

4. We have,

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\Rightarrow b^2 x^2 + a^2 y^2 = a^2 b^2$$

$$\Rightarrow b^2 r^2 \cos^2 \theta + a^2 r^2 \sin^2 \theta = a^2 b^2$$

$$\Rightarrow \frac{r^2 \cos^2 \theta}{a^2} + \frac{r^2 \sin^2 \theta}{b^2} = 1$$

$$\Rightarrow \frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2} = \frac{1}{r^2}$$

5. We have,

$$2x^2 + 3xy + 2y^2 = 1$$

$$\Rightarrow 2r^2 \cos^2 \theta + 3r^2 \sin \theta \cos \theta + 2r^2 \sin^2 \theta =$$

1

$$\Rightarrow r^2 (2 \cos^2 \theta + 3 \sin \theta \cos \theta + 2 \sin^2 \theta) = 1$$

$$\Rightarrow r^2 \left(2 + \frac{3}{2} \sin 2\theta \right) = 1$$

$$\Rightarrow r^2 = \frac{2}{(4 + 3 \sin 2\theta)}$$

6. The required distance

$$\begin{aligned} &= \sqrt{a^2(\cos \alpha - \cos \beta)^2 + a^2(\sin \alpha - \sin \beta)^2} \\ &= a\sqrt{(\cos \alpha - \cos \beta)^2 + (\sin \alpha - \sin \beta)^2} \\ &= a\sqrt{[(\cos^2 \alpha + \sin^2 \alpha) + (\cos^2 \beta + \sin^2 \beta) - 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta)]} \\ &= a\sqrt{2 - 2\cos(\alpha - \beta)} \\ &= a\sqrt{2(1 - \cos(\alpha - \beta))} \\ &= 2a \left| \sin\left(\frac{\alpha - \beta}{2}\right) \right| \end{aligned}$$

7. The required distance

$$\begin{aligned} &= \sqrt{3^2 + 7^2 - 2 \times 3 \times 7 \times \cos\left(\frac{5\pi}{4} - \frac{\pi}{4}\right)} \\ &= \sqrt{3^2 + 7^2 - 2 \times 3 \times 7 \times \cos(\pi)} \\ &= \sqrt{9 + 49 + 42} = \sqrt{100} = 10 \end{aligned}$$

8. Let $P = (x, y)$, $A = (6, -1)$ and $B = (2, 3)$.

We have,

$$\begin{aligned} PA &= PB \\ \Rightarrow (x - 6)^2 + (y + 1)^2 &= (x - 2)^2 + (y - 3)^2 \\ \Rightarrow x^2 - 12x + 36 + y^2 + 2y + 1 &= x^2 - 4x + 4 + y^2 - 6y + 9 \\ \Rightarrow -12x + 36 + 2y + 1 &= -4x + 4 - 6y + 9 \\ \Rightarrow -8x + 8y + 37 - 13 &= 0 \\ \Rightarrow -8x + 8y + 24 &= 0 \\ \Rightarrow A = -8, B = 8, C = 24 \end{aligned}$$

Hence, the value of $A + B + C + 10$
 $= -8 + 8 + 24 + 10 = 34$

9. We have,

$$AB = \sqrt{(2 + 2)^2 + (-1 - 3)^2} = 4\sqrt{2}$$

$$BC = \sqrt{(2 - 4)^2 + (-1)^2} = \sqrt{5}$$

$$\text{and } CA = \sqrt{(4 + 2)^2 + (3)^2} = \sqrt{45} = 3\sqrt{5}$$

$$\text{Thus, } \cos A = \frac{(4\sqrt{2})^2 + (3\sqrt{5})^2 - (\sqrt{5})^2}{2 \times 4\sqrt{2} \times 3\sqrt{5}}$$

$$\Rightarrow \cos A = \frac{32 + 45 - 5}{24\sqrt{10}} = \frac{72}{24\sqrt{10}} = \frac{3}{\sqrt{10}}$$

10. Do yourself.

$$11. \left(\frac{1 \pm \sqrt{3}}{2}, \frac{7 \pm 5\sqrt{3}}{2} \right)$$

12. Let the point be (x, y) .

$$\text{Thus, } x = \frac{4 + 5}{2 + 1} = 3, y = \frac{6 - 3}{2 + 1} = 1$$

Hence, the point is $(3, 1)$.

13. Let the ratio be $m : n$.

Since the point lies on y -axis, so x co-ordinate will be zero.

$$\text{Thus, } \frac{7m - 3n}{m + n} = 0$$

$$\Rightarrow 7m - 3n = 0$$

$$\Rightarrow \frac{m}{n} = \frac{3}{7}$$

14. Let the ratio be $\lambda : 1$.

$$\text{Thus, the point is } \left(\frac{1 - 2\lambda}{\lambda + 1}, \frac{3\lambda - 2}{\lambda + 1} \right).$$

Since, the point $\left(\frac{1 - 2\lambda}{\lambda + 1}, \frac{3\lambda - 2}{\lambda + 1} \right)$ lies on the line $3x + 4y = 7$, we get

$$3\left(\frac{1 - 2\lambda}{\lambda + 1}\right) + 4\left(\frac{3\lambda - 2}{\lambda + 1}\right) = 7$$

$$\Rightarrow 3(1 - 2\lambda) + 4(3\lambda - 2) = 7\lambda + 7$$

$$\Rightarrow 12\lambda - 6\lambda - 7\lambda = 8 + 7 - 3$$

$$\Rightarrow \lambda = -12$$

Hence, the ratio is $12 : 1$ externally.

15. Let $A = (1, -2)$ and $B = (-3, 4)$ and the line AB is trisected at P and Q , respectively.

Therefore P divides A and B internally in the ratio $1 : 2$ and Q divides A and B in the ratio $2 : 1$.

Thus, the co-ordinates of P and Q are $\left(-\frac{2}{3}, 0\right)$ and $\left(-\frac{5}{3}, 2\right)$.

16. Let the co-ordinates of the vertices are (x_1, y_1) , (x_2, y_2) , and (x_3, y_3) .

$$\text{Therefore, } x_1 + x_2 = 1, x_2 + x_3 = 4, x_1 + x_3 = 3$$

$$\Rightarrow x_1 + x_2 + x_3 = 4$$

$$\text{Thus, } x_1 = 0, x_2 = 1, x_3 = 3$$

$$\text{Also, } y_1 + y_2 = 1, y_2 + y_3 = 1, y_3 + y_1 = 2$$

$$\Rightarrow y_1 + y_2 + y_3 = 2$$

$$\text{Thus, } y_1 = 1, y_2 = 0, y_3 = 1$$

Hence, the vertices are $(0, 1)$, $(1, 0)$ and $(3, 1)$.

17. Let the co-ordinates of the third vertex be (x, y) .

As we know that the diagonals of a parallelogram bisect each other.

$$\text{Thus, } \frac{x - 1}{2} = \frac{1 + 2}{2} \text{ and } \frac{y + 2}{2} = \frac{5 + 3}{2}$$

$$\Rightarrow x = 4, y = 6$$

Hence, the fourth vertex is $(4, 6)$.

18. The co-ordinates of the centroid of $\triangle ABC$ are

$$\left(\frac{2 + 6 + 2}{3}, \frac{4 + 4 + 0}{3} \right) = \left(\frac{10}{3}, \frac{8}{3} \right)$$

19. Let the co-ordinates of the third vertex be (x, y) .

$$\text{Therefore, } \frac{x - 1 + 5}{3} = 0 \Rightarrow x = -4 \text{ and}$$

$$\frac{y + 4 + 2}{3} = -3 \Rightarrow y = -15$$

Hence, the co-ordinates of the third vertex be $(-4, -15)$.

20. Let $a = BC$, $b = CA$, $c = AB$ be the lengths of the side of the given $\triangle ABC$.

$$\text{Therefore, } a = \sqrt{(3-2)^2 + (4-3)^2} = \sqrt{2},$$

$$b = \sqrt{(3-1)^2 + (4-2)^2} = 2\sqrt{2}$$

$$\text{and } c = \sqrt{(2-1)^2 + (3-2)^2} = \sqrt{2}$$

Thus, the incentre are

$$\begin{aligned} & \left(\frac{ax_1 + bx_2 + cx_3}{a+b+c}, \frac{ay_1 + by_2 + cy_3}{a+b+c} \right) \\ &= \left(\frac{1 \cdot \sqrt{2} + 2 \cdot 2\sqrt{2} + 3 \cdot \sqrt{2}}{\sqrt{2} + 2\sqrt{2} + \sqrt{2}}, \frac{2 \cdot \sqrt{2} + 3 \cdot 2\sqrt{2} + 4 \cdot \sqrt{2}}{\sqrt{2} + 2\sqrt{2} + \sqrt{2}} \right) \\ &= \left(\frac{8\sqrt{2}}{4\sqrt{2}}, \frac{12\sqrt{2}}{4\sqrt{2}} \right) \\ &= (2, 3) \end{aligned}$$

21. As we know that the centroid divides the orthocentre and the circumcentre in the ratio 2 : 1.

Thus, the centroid are

$$\left(\frac{2 \cdot \frac{3}{2} + 1 \cdot 1}{2+1}, \frac{2 \cdot \frac{3}{4} + 1 \cdot 1}{2+1} \right) = \left(\frac{4}{3}, \frac{5}{6} \right)$$

22. Clearly, it is a right-angled triangle.

As we know that in case of a right angled triangle, the circumcentre is the mid-point of the hypotenuse and the orthocentre is at right angle.

$$\begin{aligned} \text{Thus, Circumcentre} &= \left(\frac{0+2}{2}, \frac{2+0}{2} \right) \\ &= (1, 1) \end{aligned}$$

and

$$\text{Orthocentre} = (0, 0)$$

$$\begin{aligned} \text{Hence, the required distance} &= \sqrt{(1-0)^2 + (1-0)^2} \\ &= \sqrt{2} \end{aligned}$$

23. The required area

$$\begin{aligned} &= \frac{1}{2} \begin{vmatrix} x_1 - x_3 & x_2 - x_3 \\ y_1 - y_3 & y_2 - y_3 \end{vmatrix} \\ &= \frac{1}{2} \begin{vmatrix} 3-7 & 7+1 \\ -4-5 & 5-10 \end{vmatrix} \\ &= \frac{1}{2} \begin{vmatrix} -4 & 8 \\ -9 & -5 \end{vmatrix} \\ &= \frac{1}{2} (20 + 72) = 46 \text{ sq.u.} \end{aligned}$$

24. The required area

$$\begin{aligned} &= \frac{1}{2} \begin{vmatrix} x_1 - x_3 & x_2 - x_3 \\ y_1 - y_3 & y_2 - y_3 \end{vmatrix} \\ &= \frac{1}{2} \begin{vmatrix} t-t-3 & t+3-t-2 \\ t+2-t & t-t-2 \end{vmatrix} \end{aligned}$$

$$= \frac{1}{2} \begin{vmatrix} -3 & 1 \\ 2 & -2 \end{vmatrix}$$

$$= \frac{1}{2} (6 - 2) = 2 \text{ sq.u.}$$

25. Given area of a triangle is 6.

$$\Rightarrow \frac{1}{2} \begin{vmatrix} x-1 & 1-2 \\ y-2 & 2-1 \end{vmatrix} = \pm 6$$

$$\Rightarrow \begin{vmatrix} x-1 & -1 \\ y-2 & 1 \end{vmatrix} = \pm 12$$

$$\Rightarrow (x-1) + (y-2) = \pm 12$$

$$\Rightarrow x + y - 3 + 12 = 0, x + y - 3 - 12 = 0$$

$$\Rightarrow x + y + 9 = 0, x + y - 15 = 0$$

Hence, the result.

26. The required area of a quadrilateral

$$= \frac{1}{2} \begin{vmatrix} 1 & 1 \\ 7 & -3 \\ 12 & 2 \\ 7 & 21 \\ 1 & 1 \end{vmatrix}$$

$$= \frac{1}{2} \{(-3 + 14 + 252 + 7) - (21 + 14 - 36 + 7)\}$$

$$= \frac{1}{2} (270 - 6) = 132 \text{ sq.u.}$$

27. The required area of a pentagon

$$= \frac{1}{2} \begin{vmatrix} 4 & 3 \\ -5 & 6 \\ 0 & 7 \\ 3 & -6 \\ -7 & -2 \\ 4 & 3 \end{vmatrix}$$

$$= \frac{1}{2} \{(24 - 35 + 0 - 6 - 21) - (-8 + 42 + 21 + 0 + 15)\}$$

$$= \frac{1}{2} (38 + 70) = 54 \text{ sq. u.}$$

28. We have,

$$\frac{1}{2} \begin{vmatrix} x_1 - x_2 & x_2 - x_3 \\ y_1 - y_2 & y_2 - y_3 \end{vmatrix}$$

$$= \frac{1}{2} \begin{vmatrix} a-b & b-c \\ b+c-c-a & c+a-a-b \end{vmatrix}$$

$$= \frac{1}{2} \begin{vmatrix} a-b & b-c \\ b-a & c-b \end{vmatrix}$$

$$= \frac{1}{2} (a-b)(b-c) \begin{vmatrix} 1 & 1 \\ -1 & -1 \end{vmatrix}$$

$$= 0$$

Hence, the result.

29. Now,

$$\begin{aligned} \text{ar}(\triangle DBC) &= \frac{1}{2} \begin{vmatrix} x+3 & -3-4 \\ 3x-3 & 5+2 \end{vmatrix} \\ &= \frac{1}{2} (7x+21-21x-21) \\ &= \frac{1}{2} (-14x) = 7x \end{aligned}$$

and

$$\begin{aligned} \text{ar}(\triangle ABC) &= \frac{1}{2} \begin{vmatrix} 6+3 & -3-4 \\ 3-5 & 5+2 \end{vmatrix} \\ &= \frac{1}{2} \begin{vmatrix} 9 & -7 \\ -2 & 7 \end{vmatrix} \\ &= \frac{1}{2} (63-14) = \frac{49}{2} \end{aligned}$$

It is given that

$$\frac{7x}{\frac{49}{2}} = \frac{1}{2}$$

$$\Rightarrow x = \frac{7}{4}$$

30. The required area

$$= \frac{1}{2 \times |C_1 C_2 C_3|} \begin{vmatrix} 7 & -2 & 10 \\ 7 & 2 & -10 \\ 9 & 1 & 2 \end{vmatrix},$$

$$\text{where } C_1 = \begin{vmatrix} 7 & 2 \\ 9 & 1 \end{vmatrix} = -11, \quad C_2 = -\begin{vmatrix} 7 & -2 \\ 9 & 1 \end{vmatrix} = -25,$$

$$\text{and } C_3 = -\begin{vmatrix} 7 & -2 \\ 7 & 2 \end{vmatrix} = -28$$

$$= \frac{1}{2 \times 11 \times 25 \times 28} (96)^2$$

$$= \frac{9216}{2 \times 11 \times 25 \times 28}$$

$$= \frac{9216}{15400} = \frac{2304}{3850} \text{ sq. u.}$$

31. Let $x = a \cos \theta$, $y = a \sin \theta$

$$\text{Then } x^2 + y^2 = a^2 \cos^2 \theta + a^2 \sin^2 \theta = a^2$$

which is the required locus of the point P .

32. Let $x = at^2$ and $y = 2at$.

$$\text{Then, } y^2 = 4a^2 t^2 = 4a(at^2) = 4ax$$

which is the required locus of P .

33. Let the movable point P be (x, y) and the point on y -axis be $(0, y)$.

Given condition is

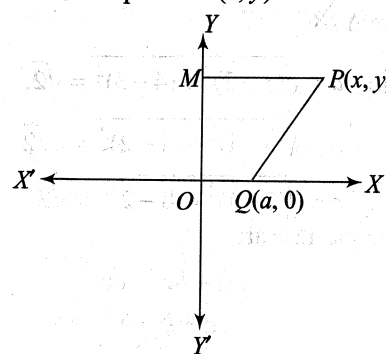
$$\sqrt{x^2 + y^2} = 3x$$

$$\Rightarrow x^2 + y^2 = 9x^2$$

$$\Rightarrow y^2 = 8x^2$$

which is the required locus of P .

34. Let the movable point be (x, y) .



Given condition is

$$PQ = PM$$

$$\Rightarrow \sqrt{(x-a)^2 + y^2} = x$$

$$\Rightarrow (x-a)^2 + y^2 = x^2$$

$$\Rightarrow y^2 = x^2 - (x-a)^2 = 2ax - a^2$$

which is the required locus of the given movable point.

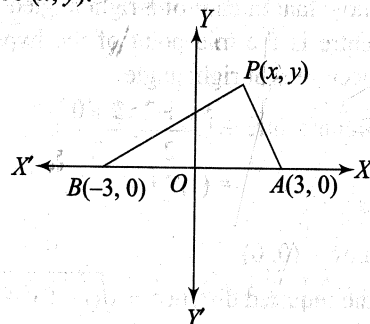
35. Let the co-ordinates of the variable point P be (x, y) .

$$\text{Then } x = t + \frac{1}{t} \text{ and } y = t - \frac{1}{t}$$

$$\Rightarrow x^2 - y^2 = \left(t + \frac{1}{t}\right)^2 - \left(t - \frac{1}{t}\right)^2 = 4$$

which is the required locus of P and represents a rectangular hyperbola.

36. Let P be (x, y) .



Given condition is

$$PA + PB = 8.$$

$$\Rightarrow \sqrt{(x-3)^2 + y^2} + \sqrt{(x+3)^2 + y^2} = 8$$

$$\Rightarrow \sqrt{(x-3)^2 + y^2} = 8 - \sqrt{(x+3)^2 + y^2}$$

$$\Rightarrow (x-3)^2 + y^2 = 64 + ((x+3)^2 + y^2) - 16\sqrt{(x-3)^2 + y^2}$$

$$\Rightarrow 6y = 64 - 6y - 16\sqrt{(x-3)^2 + y^2}$$

$$\Rightarrow 12y - 64 = 16\sqrt{(x-3)^2 + y^2}$$

$$\Rightarrow 3y - 16 = 4\sqrt{(x-3)^2 + y^2}$$

$$\Rightarrow (3y - 16)^2 = 16((x-3)^2 + y^2)$$

$$\Rightarrow 9y^2 - 96y + 256 = 16(x^2 + y^2 - 6x + 9)$$

$$\Rightarrow 16x^2 + 7y^2 = 256 - 144 = 112$$

$$\Rightarrow \frac{x^2}{7} + \frac{y^2}{16} = 1$$

which is the required locus of the given point and represents an ellipse.

37. Let $AB = l$.

Consider the point A lies on x -axis and B lies on y -axis such that

$$A = (a, 0) \text{ and}$$

$$B = (0, b)$$

Therefore, $AB = l$

$$\Rightarrow \sqrt{a^2 + b^2} = l$$

$$\Rightarrow a^2 + b^2 = l^2$$

...(i)

Let the mid-point of A and B be (α, β)

$$\text{Thus, } \alpha = \frac{a}{2}, \beta = \frac{b}{2}$$

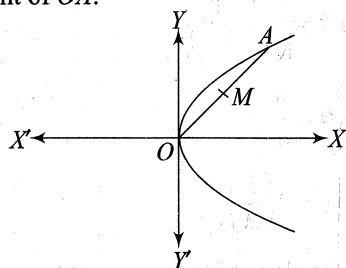
$$\Rightarrow a = 2\alpha, b = 2\beta$$

From Eq. (i), we get,

$$\alpha^2 + \beta^2 = \frac{l^2}{4}$$

Hence, the locus of (α, β) is $x^2 + y^2 = \frac{l^2}{4}$.

38. Let the point M be (α, β) and $A(2\alpha, 2\beta)$ where M is the mid-point of OA .



Since A lies on the curve $y^2 = 8x$, so

$$(2\beta)^2 = 8(2\alpha)$$

$$\Rightarrow 4\beta^2 = 16\alpha$$

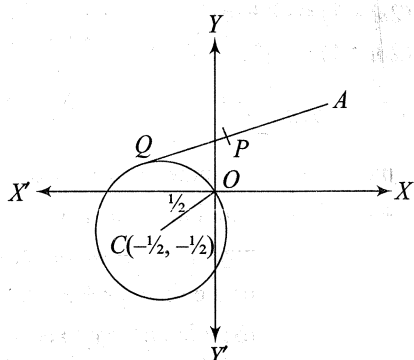
$$\Rightarrow \beta^2 = 4\alpha$$

Hence, the required locus of M is $y^2 = 4x$.

39. We have

$$x^2 + y^2 + x + y = 0$$

$$\Rightarrow \left(x + \frac{1}{2}\right)^2 + \left(y + \frac{1}{2}\right)^2 = \left(\frac{1}{\sqrt{2}}\right)^2$$



Let the point Q be

$$\left(-\frac{1}{2} + \frac{1}{\sqrt{2}} \cos \theta, -\frac{1}{2} + \frac{1}{\sqrt{2}} \sin \theta\right)$$

and P be (α, β)

$$\text{Then } \alpha = \frac{1}{4} + \frac{1}{2\sqrt{2}} \cos \theta, \beta = \frac{3}{4} + \frac{1}{2\sqrt{2}} \sin \theta$$

$$\begin{aligned} \text{Now, } 8\left(\alpha - \frac{1}{4}\right)^2 + 8\left(\beta - \frac{3}{4}\right)^2 \\ = \cos^2 \theta + \sin^2 \theta = 1 \end{aligned}$$

$$\Rightarrow 8\left(\alpha^2 + \beta^2 - \frac{\alpha}{2} - \frac{3\beta}{2}\right) + \frac{1}{2} + \frac{9}{2} - 1 = 0$$

$$\Rightarrow 8\left(\alpha^2 + \beta^2 - \frac{\alpha}{2} - \frac{3\beta}{2}\right) + 4 = 0$$

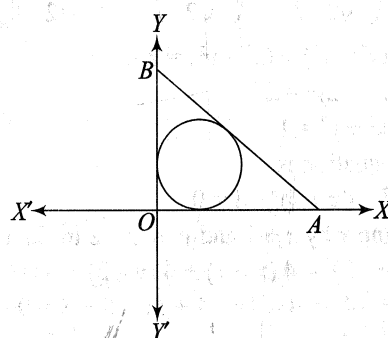
$$\Rightarrow 2(\alpha^2 + \beta^2) - \alpha - 3\beta + 1 = 0$$

Hence, the locus of (α, β) is

$$2(x^2 + y^2) - x - y + 1 = 0,$$

40. Let $OA = a$ and $OB = b$.

Since the circle touches both the axes and the line AB , so the inradius of $\triangle OAB$ be 2.



From trigonometry, we can write

$$r = \frac{\Delta}{s}, \quad \dots(i)$$

where Δ = area of a triangle

and s = semi-perimeter of the triangle.

Let (x, y) be the circumcentre of $\triangle OAB$, then

$$x = \frac{a}{2} \text{ and } y = \frac{b}{2}$$

Thus, $a = 2x$ and $b = 2y$.

Putting the values of a and b in Eq. (i), we get

$$2 = \frac{\frac{1}{2} \cdot (2x) \cdot (2y)}{\left(\frac{2x + 2y + \sqrt{4x^2 + 4y^2}}{2}\right)}$$

$$\Rightarrow xy = x + y + \sqrt{x^2 + y^2}$$

is the required locus.

$$44. x^2 + y^2 = 4$$

$$45. x^2 + y^2 = 3$$

46. Given curve is

$$2x^2 + y^2 - 3x + 5y - 8 = 0 \quad (i)$$

Replacing x by $x - 1$ and y by $y + 2$ in Eq. (i), we get

$$2(x - 1)^2 + (y + 2)^2 - 3(x - 1) + 5(y + 2) - 8 = 0$$

$$\Rightarrow 2(x^2 - 2x + 1) + (y^2 + 4y + 4)$$

$$-3(x - 1) + 5(y + 2) - 8 = 0$$

$$\Rightarrow 2x^2 + y^2 - 7x + 9y + 11 = 0$$

47. Given curve is
 $x^2 + y^2 = 36$... (i)
 Replacing x by $x - 4$ and y by $y - 5$ in Eq. (i), we get
 $(x - 4)^2 + (y - 5)^2 = 36$
 $\Rightarrow x^2 + y^2 - 8x - 10y + 16 + 25 - 36 = 0$
 $\Rightarrow x^2 + y^2 - 8x - 10y + 5 = 0$
 which is the required equation of the original axes.
48. Let the point be (x, y) .
 Thus $x - 1 = -7, y + 8 = 3$
 $\Rightarrow x = -6, y = -5$
 Hence, the point be $(-6, -5)$.
49. Given equation is
 $3x^2 + 2xy + 3y^2 = 2$ (i)
 Replacing x by $(x \cos 45^\circ - y \sin 45^\circ)$
 and y by $(x \cos 45^\circ + y \sin 45^\circ)$ in Eq. (i), we get
 $3\left(\frac{x-y}{\sqrt{2}}\right)^2 + 2\left(\frac{x-y}{\sqrt{2}}\right)\left(\frac{x+y}{\sqrt{2}}\right) + 3\left(\frac{x+y}{\sqrt{2}}\right)^2 = 2$
 $\Rightarrow 3(x^2 + y^2) + (x^2 - y^2) = 2$
 $\Rightarrow 4x^2 + 2y^2 = 2$
 $\Rightarrow 2x^2 + y^2 = 1$
50. Given equation is
 $y^2 - 4x + 4y + 8 = 0$ (i)
 Replacing x by $x + 1$ and y by $y - 2$ in Eq. (i), we get
 $(y - 2)^2 - 4(x + 1) + 4(y - 2) + 8 = 0$
 $\Rightarrow y^2 - 4y + 4 - 4x - 4 + 4y - 8 + 8 = 0$
 $\Rightarrow y^2 = 4x$
 which is the required transformed equation.
51. If the point $P(1, 2)$ is translated itself 2 units along the positive direction of x -axis, then P becomes $(3, 2)$.
 Let $Z = (3, 2) = 3 + 2i$
 Thus, the new position of $P = iZ$
 $= i(3 + 2i)$
 $= -2 + 3i = (-2, 3)$
52. 12
53. $x^2 - y^2 = a^2$
54. $4x^2 + 2y^2 = 1$
55. $(-3, 2)$
56. $\left(\frac{1-4\sqrt{3}}{2}, \frac{4+\sqrt{3}}{2}\right)$

LEVEL I

(STRAIGHT LINE)

1. Hence, the required slope of $PQ = \frac{y_2 - y_1}{x_2 - x_1} = \frac{10 - 4}{3 - 2} = 6$
2. Given slope = 2
 $\Rightarrow \frac{7 - 5}{\lambda - 2} = 2$
 $\Rightarrow \lambda - 2 = 1$
 $\Rightarrow \lambda = 3$

3. Let m_1 be the line joining the points $(2, -3)$ and $(-5, 1)$ and m_2 be the line joining the points $(7, -1)$ and $(0, 3)$.

$$\text{Thus } m_1 = \frac{1+3}{-5-2} = -\frac{4}{7} \text{ and } m_2 = \frac{3+1}{0-7} = -\frac{4}{7}.$$

Since, $m_1 = m_2$, so the slopes are parallel.

4. Consider the points $P = (a, b + c)$,

$$Q = (b, c + a), R = (c, a + b)$$

$$\text{Therefore, } m(PQ) = \frac{c + a - b - c}{b - a} = \frac{a - b}{b - a} = -1$$

$$\text{and } m(QR) = \frac{a + b - c - a}{c - b} = \frac{b - c}{c - b} = -1$$

$$\text{Since, } m(PQ) = m(QR)$$

$\Rightarrow$ Thus, the points P, Q and R are collinear.

5. Let m_1 and m_2 be the slopes of the line joining the points $(0, 0), (2, 2)$ and $(2, -2), (3, 5)$.

$$\text{Therefore, } m_1 = \frac{2-0}{2-0} = 1 \text{ and } m_2 = \frac{5+2}{3-2} = 7$$

Let θ be the angle between them

$$\tan \theta = \left| \frac{7-1}{1+7 \cdot 1} \right|$$

$$\Rightarrow \tan \theta = \frac{3}{4}$$

$$\Rightarrow \theta = \tan^{-1}\left(\frac{3}{4}\right)$$

6. Let m be the slope of the other line.

Given,

$$\tan 45^\circ = \left| \frac{m - \frac{1}{2}}{1 + \frac{m}{2}} \right|$$

$$\Rightarrow \tan 45^\circ = \left| \frac{2m - 1}{2 + m} \right|$$

$$\Rightarrow (2m - 1) = \pm (2 + m)$$

$$\Rightarrow (2m - 1) = (2 + m)$$

$$\text{and } (2m - 1) = -(2 + m)$$

$$\Rightarrow m = 3 \text{ and } m = -\frac{1}{3}$$

Hence, the slopes of the other line are 3 or $-1/3$.

7. Equation of a line parallel to x -axis is $y = k$.

Which is passing through $(2, -3)$, we have, $k = -3$.

Hence, the equation of the line is $y + 3 = 0$.

8. Equation of a line perpendicular to y -axis is $y = k$ which is passing through the point $(3, 4)$, we have, $k = 4$.

Hence, the equation of the line is $y = 4$.

9. Hence, the equation of a line, which is equidistant from the lines $x = 6$ and $x = 10$ is $x = \frac{6+10}{2} = 8$

10. Given $m = 3$ and $c = 7$.

Hence, the required equation of the line is

$$y = mx + c = 3x + 7$$

11. Here, $m = \tan(135^\circ) = -1$ and $c = 5$.

Hence, the required equation of the given line is

$$y = mx + c = -x + 5$$

12. Given,

$$\theta = \tan^{-1}\left(\frac{3}{5}\right)$$

$$\Rightarrow \tan \theta = \frac{3}{5}$$

$$\Rightarrow m = \frac{3}{5} \text{ and } c = -6$$

Hence, the equation of the given straight line is

$$y = mx + c = \frac{3}{5}x - 6$$

13. Given $m = \pm 1$ and $c = 4$.

Hence, the equation of the line is

$$y = mx + c = \pm x + 4$$

14. Here, $m = \tan(45^\circ) = 1$

and $m = \tan(135^\circ) = -1$ and $c = 0$

Hence, the equation of the bisectors is

$$y = mx + c = \pm x$$

15. Here, $m = \tan(120^\circ) = -\sqrt{3}$ and the point $(x_1, y_1) = (2, 3)$

Hence, the equation of a straight line is

$$y - y_1 = m(x - x_1)$$

$$\Rightarrow y - 3 = -\sqrt{3}(x - 2)$$

$$\Rightarrow \sqrt{3}x + y = 2\sqrt{3} + 3$$

16. Let the points A and B are $(1, 2)$ and $(5, 7)$, respectively.

$$\text{Therefore, } m(AB) = \frac{7-2}{5-1} = \frac{5}{4}$$

and the mid-point of A and B are

$$\left(\frac{1+5}{2}, \frac{2+7}{2}\right) = \left(3, \frac{9}{2}\right)$$

Hence, the equation of the right (perpendicular) bisector is

$$y - \frac{9}{2} = -\frac{4}{5}(x - 3)$$

$$\Rightarrow 8x + 10y = 69$$

17. Given,

$$\cos \theta = -\frac{3}{5}$$

$$\Rightarrow \tan \theta = -\frac{4}{5}$$

$$\Rightarrow m = -\frac{4}{5}$$

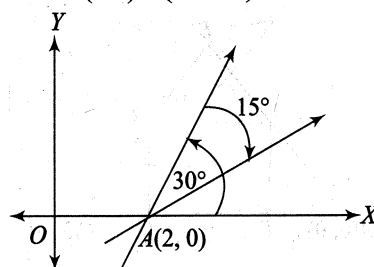
Hence, the equation of the given line is

$$y - 2 = -\frac{4}{5}(x - 1)$$

$$\Rightarrow 4x + 5y = 14$$

18. We have,

$$m = \tan(15^\circ) = (2 - \sqrt{3})$$



Hence, the required equation of the line is

$$y - 0 = (2 - \sqrt{3})(x - 2)$$

$$\Rightarrow (2 - \sqrt{3})x - y - 2(2 - \sqrt{3}) = 0$$

19. Hence, the equation of a line is

$$\begin{vmatrix} x - x_1 & y - y_1 \\ x_1 - x_2 & y_1 - y_2 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x - 1 & y - 2 \\ 3 - 1 & 4 - 2 \end{vmatrix} = 0$$

$$\Rightarrow 2(x - 1) - 2(y - 2) = 0$$

$$\Rightarrow x - y + 1 = 0.$$

20. Here slope of $BC = \frac{-1-9}{-2+4} = -\frac{10}{2} = -5$

Hence, the equation of the altitude through A is

$$y - 4 = \frac{1}{5}(x - 10)$$

$$\Rightarrow x - 5y + 10 = 0$$

21. The co-ordinates of the mid points of B and C are

$$\left(\frac{2+5}{2}, \frac{3+4}{2}\right) = \left(\frac{7}{2}, \frac{7}{2}\right)$$

Hence, the required equation of the median through A

$$\text{is } \begin{vmatrix} x - 1 & y - 2 \\ \frac{7}{2} - 1 & \frac{7}{2} - 2 \end{vmatrix} = 0$$

$$\Rightarrow 3(x - 1) - 5(y - 2) = 0$$

$$\Rightarrow 3x - 5y + 7 = 0$$

22. Here, $AB = \sqrt{10}$, $AC = \sqrt{10}$, $BC = 2\sqrt{10}$

Let AD is the internal bisector of the angle $\angle BAC$.

If AD is the internal bisector of the angle $\angle BAC$, then

$$\frac{BD}{DC} = \frac{AB}{AC} = \frac{\sqrt{10}}{\sqrt{10}} = 1$$

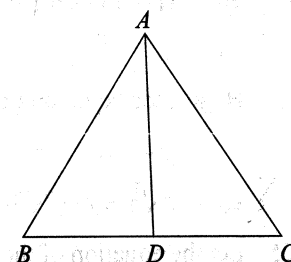
$$\Rightarrow BD : DC = 1 : 1$$

Thus D is the mid-point of B and C , i.e. $D = (4, 4)$

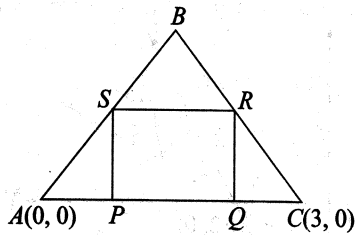
Hence, the equation of the internal bisector is

$$y - 2 = \frac{4-2}{4-5}(x - 5) = -2(x - 5)$$

$$\Rightarrow 2x + y = 12$$



23. Let the square be PQRS, whose length of the side is k . Consider the point of P be $(a, 0)$.



Then $Q = (a + k, 0)$, $R = (a + k, k)$ and $S = (a, k)$.
The equation of the line AB is

$$y - 0 = \frac{1 - 0}{2 - 0}(x - 0)$$

$$\Rightarrow x - 2y = 0 \quad \dots(i)$$

and the equation of the line BC is

$$y - 1 = \frac{0 - 1}{3 - 2}(x - 2)$$

$$\Rightarrow x + y = 3 \quad \dots(ii)$$

Since, the point $S(a, k)$ on Eq. (i), we get,

$$a = 2k$$

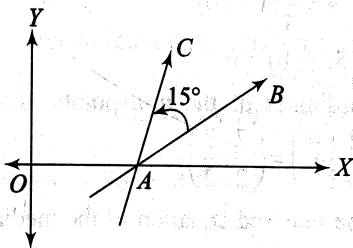
and the point $R(a + k, k)$ on Eq. (ii), we get,
 $a + 2k = 3$.

Thus, $k = 3/4$ and $a = 3/2$.

Hence, the co-ordinates of P, Q, R and S are

$(3/2, 0)$, $(9/4, 0)$, $(9/4, 3/4)$ and $(3/2, 3/4)$.

24. Let slope of AC is m . The slope of AB is 1. We have



$$\tan(15^\circ) = \left| \frac{m - 1}{1 + m} \right|$$

$$\Rightarrow (2 - \sqrt{3})m + (2 - \sqrt{3}) = m - 1$$

$$\Rightarrow (1 - \sqrt{3})m = (-3 + \sqrt{3})$$

$$\Rightarrow (1 - \sqrt{3})m = (-3 + \sqrt{3}) = \sqrt{3}(1 - \sqrt{3})$$

$$\Rightarrow m = \sqrt{3}$$

Hence, the equation of the line AC is

$$y - 0 = \sqrt{3}(x - 2)$$

$$\Rightarrow x\sqrt{3} - y - 2\sqrt{3} = 0$$

25. Let the equation of the line is $\frac{x}{a} + \frac{y}{b} = 1$.

Since, the line makes equal intercepts with the axes, so $a = b$.

Thus $x + y = a$ which is passing through $(3, 4)$.

Therefore $a = 7$.

Hence, the equation of the required line is $x + y = 7$.

26. Let the equation of the line is $\frac{x}{a} + \frac{y}{b} = 1$... (i)

Given condition is $a = 2b$... (ii)

From Eqs (i) and (ii), we get,

$$2x + y = 2b$$

Which is passing through the point $(2, 3)$.

Therefore, $2b = 4 + 3 = 7$

$$\Rightarrow b = \frac{7}{2}$$

Hence, the equation of the line is $2x + y = 7$.

27. Let the equation of the line be $\frac{x}{a} + \frac{y}{b} = 1$,

where $A = (a, 0)$ and $B = (0, b)$.

Consider the given line is bisected at the point $M(2, 3)$.

$$\text{Thus, } 2 = \frac{a}{2} \Rightarrow a = 4$$

$$\text{and } 3 = \frac{b}{2} \Rightarrow b = 6$$

Hence, the equation of the line is

$$\frac{x}{4} + \frac{y}{6} = 1$$

$$\Rightarrow 3x + 2y = 12$$

28. Let the equation of the line be $\frac{x}{a} + \frac{y}{b} = 1$, where $A = (a, 0)$ and $B = (0, b)$

Given condition is $a = -b$

Therefore, $x - y = a$ which is passing through $(3, -4)$

Thus, $a = 7$.

Hence, the equation of the line is $x - y = 7$.

29. Let the equation of the line is $\frac{x}{a} + \frac{y}{b} = 1$

which is passing through $(3, 2)$, so

$$\frac{3}{a} + \frac{2}{b} = 1 \quad \dots(i)$$

Also given condition is $a - b = 2$... (ii)

From Eq. (i) and (ii), we get

$$3b + 2a = ab = b(b + 2)$$

$$\Rightarrow 3b + 2b + 4 = b^2 + 2b$$

$$\Rightarrow b^2 - 3b - 4 = 0$$

$$\Rightarrow (b - 4)(b + 1) = 0$$

$$\Rightarrow b = -1, 4$$

Thus, $a = 1, 6$.

Hence, the equation of the line are

$$x - y = 1, \text{ and } 2x - 3y = 12.$$

30. Let the equation of the line AB is

$$2x + 3y = 6 \Rightarrow \frac{x}{3} + \frac{y}{2} = 1$$

Given lines are $x, y = 0$

$$\Rightarrow x = 0, y = 0$$

Hence, the area of a triangle $= \frac{1}{2} \cdot 3 \cdot 2 = 3$ sq. u.

$$31. 2x + y = 10.$$

$$32. 3x - 8y = 0, 3x - 2y = 0.$$

33. Do yourself

34. Do yourself.

35. Here, $p = 5$ and $\alpha = 60^\circ$

Hence, the equation of the line is

$$x \cos \alpha + y \sin \alpha = p$$

$$\Rightarrow x \cos (60^\circ) + y \sin (60^\circ) = 5$$

$$\Rightarrow x + \sqrt{3}y = 10$$

36. Here, $p = 2$ and $\tan \theta = \frac{5}{12}$

$$\Rightarrow \sin \theta = \frac{5}{13} \text{ and } \cos \theta = \frac{12}{13}$$

Hence, the equation of the line is

$$x \cos \theta + y \sin \theta = p$$

$$\Rightarrow 12x + 5y = 26.$$

37. Hence, the co-ordinates of the required points are

$$(x_1 \pm r \cos \theta, y_1 \pm r \sin \theta).$$

Here $r = 5$, $(x_1, y_1) = (3, 2)$

$$\text{and } \tan \theta = \frac{3}{4} \Rightarrow \cos \theta = \frac{4}{5}, \sin \theta = \frac{3}{5}$$

Thus, the points are

$$(x_1 \pm r \cos \theta, y_1 \pm r \sin \theta)$$

$$= \left(3 \pm 5 \cdot \frac{4}{5}, 2 \pm 5 \cdot \frac{3}{5} \right)$$

$$= (3 \pm 4, 2 \pm 3)$$

$$= (7, 5), (-1, -1)$$

38. Here, $r = 4\sqrt{2}$, $(x_1, y_1) = (-2, 3)$ and $\theta = 45^\circ$.

Hence, the co-ordinates of the points are

$$(x_1 \pm r \cos \theta, y_1 \pm r \sin \theta)$$

$$= \left(-2 \pm 4\sqrt{2} \cdot \frac{1}{\sqrt{2}}, 3 \pm 4\sqrt{2} \cdot \frac{1}{\sqrt{2}} \right)$$

$$= (-2 \pm 4, 3 \pm 4)$$

$$= (2, 7) \text{ and } (-6, -1)$$

39. Given point P is $(3, 4)$.

Here, $r = \sqrt{2}$ and $\theta = 45^\circ$

Let Q be the new position of P .

Hence, the co-ordinates of Q are

$$\left(3 + \sqrt{2} \cdot \frac{1}{\sqrt{2}}, 4 + \sqrt{2} \cdot \frac{1}{\sqrt{2}} \right) = (4, 5)$$

40. Let the new position of B is C .

Here, $r = AB = AC = \sqrt{(3-2)^2 + (1-0)^2} = \sqrt{2}$ and

$$\theta = 45^\circ + 15^\circ = 60^\circ$$

Thus, the co-ordinates of C are

$$(2 + \sqrt{2} \cos(60^\circ), 0 + \sqrt{2} \sin(60^\circ))$$

$$= \left(2 + \sqrt{2} \cdot \frac{1}{2}, \sqrt{2} \cdot \frac{\sqrt{3}}{2} \right) = \left(2 + \frac{1}{\sqrt{2}}, \sqrt{\frac{3}{2}} \right)$$

41. Let the point P be $(1, 2)$ and the line PQ makes an angle θ with the positive direction of x -axis.

Here, $r = \sqrt{2}$ and $(x_1, y_1) = (1, 2)$

Thus, the co-ordinates of Q are

$$\left(1 + \sqrt{\frac{2}{3}} \cos \theta, 2 + \sqrt{\frac{2}{3}} \sin \theta \right)$$

Since, the point Q lies on $x + y = 4$, so

$$1 + \sqrt{\frac{2}{3}} \cos \theta + 2 + \sqrt{\frac{2}{3}} \sin \theta = 4$$

$$\Rightarrow \sqrt{\frac{2}{3}} \cos \theta + \sqrt{\frac{2}{3}} \sin \theta = 1$$

$$\Rightarrow \cos \theta + \sin \theta = \sqrt{\frac{3}{2}}$$

$$\Rightarrow \frac{1}{\sqrt{2}} \cos \theta + \frac{1}{\sqrt{2}} \sin \theta = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \left(\theta - \frac{\pi}{4} \right) = \pm \left(\frac{\pi}{6} \right)$$

$$\Rightarrow \theta = \frac{\pi}{4} \pm \frac{\pi}{6} = 75^\circ, 15^\circ.$$

42. $R(-2, -1)$, $Q(-1, 2)$ and $S(1, -2)$

43. $\left(-\frac{3}{2}, \frac{3}{2} \right)$ and $\left(\frac{1}{2}, -\frac{3}{2} \right)$

44. $\sqrt{2}$

45. 6

46. $4\sqrt{2}$

47. $\sqrt{5}$

48. 5.

49. $(8, 2)$ and $(0, -4)$

50. Given line is $x + \sqrt{3}y + 4 = 0$

(i) slope intercept form is

$$\Rightarrow y = \left(-\frac{1}{\sqrt{3}} \right) x + \left(-\frac{4}{\sqrt{3}} \right)$$

$$\text{Thus, slope} = -\frac{1}{\sqrt{3}}$$

$$\text{and y-intercept} = -\frac{4}{\sqrt{3}}$$

(ii) Intercept form of $x + \sqrt{3}y + 4 = 0$ is

$$\frac{x}{(-4)} + \frac{y}{\left(-\frac{4}{\sqrt{3}} \right)} = 1$$

where x -intercept = 4 and

$$y\text{-intercept} = 4/\sqrt{3}.$$

(iii) Normal form of $x + \sqrt{3}y + 4 = 0$ is

$$-x - \sqrt{3}y = 4$$

$$\Rightarrow \left(-\frac{1}{4} \right) x + \left(\frac{-\sqrt{3}}{4} \right) y = 1$$

$$\Rightarrow x \cos \alpha + y \sin \alpha = 1,$$

$$\text{where } \alpha = \pi + \tan^{-1}(\sqrt{3}) = 240^\circ$$

51. (i) Let $(x_1, y_1) = (2, 2)$ and $(x_2, y_2) = (3, 5)$ and $a = 2$, $b = 3$, $c = 4$.

$$\text{Now, } \frac{ax_1 + by_1 + c}{ax_2 + by_2 + c} = \frac{2.2 + 3.2 + 4}{2.3 + 3.5 + 4} = \frac{14}{25} > 0$$

Thus, the points $(2, 2)$ and $(3, 5)$ lie on the same side of the line $2x + 3y + 4 = 0$.

(ii) Do yourself

(iii) Do yourself.

52. As we know that, if $ax_1 + by_1 + c$ and c have the same sign, then the point (x_1, y_1) lies on the origin side of the line $ax_1 + by_1 + c$.

Here, $2 > 0$ and

$$ax_1 + by_1 + c = 2.2 - 7 + 2 = 6 - 7 = -1 < 0$$

Thus, the point $(2, -7)$ does not lie on the origin side of the line $2x + y + 2 = 0$.

53. Do yourself.

54. Equation of a line parallel to

$$3x + 4y + 5 = 0 \text{ is } 3x + 4y + k = 0.$$

which passes through $(2, 3)$.

Therefore, $6 + 12 + k = 0$

$$\Rightarrow k = -18$$

Hence, the equation of the line is

$$3x + 4y - 18 = 0.$$

- 55 The co-ordinates of the mid-point of the line joining the points $(2, 3)$ and $(4, -1)$ is $(3, 1)$.

Equation of any line parallel to $3x - 4y + 6 = 0$ is

$$3x - 4y + k = 0$$

which is passing through $(3, 1)$.

Therefore, $9 - 4 + k = 0$

$$\Rightarrow k = -5$$

Hence, the equation of the line is

$$3x - 4y - 5 = 0$$

56. The slope of the line joining the points $(2, 3)$ and $(3, -1)$ is $\frac{-1-3}{3-2} = -4$

Equation of the line parallel to the given line and passing through $(2, 1)$ is

$$y - y_1 = m(x - x_1)$$

$$\Rightarrow y - 1 = -4(x - 2)$$

$$\Rightarrow 4x + y = 9$$

57. Do yourself.

58. Equation of any line perpendicular to

$$2x + 3y - 2012 = 0 \text{ is}$$

$$3x - 2y + k = 0$$

which is passing through $(3, 4)$.

Therefore, $9 - 8 + k = 0$

$$\Rightarrow k = -1$$

Hence, the equation of the line is

$$3x - 2y - 1 = 0.$$

59. Equation of any line perpendicular to $2x - 3y - 5 = 0$ is $3x + 2y - k = 0$ which is passing through $(1, 0)$.

Therefore, $3 + 0 + k = 0$

$$\Rightarrow k = -3$$

Hence, the equation of the line is

$$3x + 2y + 3 = 0.$$

60. The slope of the line joining the points $(1, 2)$ and $(3, 5)$ is $\frac{5-2}{3-1} = \frac{3}{2}$.

The mid-point of the line joining the points $(1, 2)$ and

$$(3, 5) \text{ is } \left(\frac{1+3}{2}, \frac{2+5}{2}\right) = \left(2, \frac{7}{2}\right).$$

Hence, the equation of the line is

$$y - \frac{7}{2} = -\frac{2}{3}(x - 2)$$

$$\Rightarrow 4x + 6y = 29$$

61. Slope of $BC = \frac{8-5}{3-4} = -3$

Let the altitude through A is AD .

Therefore, the slope of AD is $\frac{1}{3}$.

Hence, the equation of the altitude through A is

$$y - 2 = \frac{1}{3}(x - 1)$$

$$\Rightarrow x - 3y + 5 = 0$$

62. The point of intersection of $2x + y = 8$ and $x - y = 10$ is $(6, -4)$.

Equation of any line perpendicular to

$$3x + 4y + 2012 = 0 \text{ is}$$

$$4x - 3y + k = 0$$

which is passing through $(6, -4)$.

Therefore, $24 + 12 + k = 0 \Rightarrow k = -36$

Hence, the equation of the line is

$$4x - 3y - 36 = 0.$$

63. Hence, the distance of the point $(4, 5)$ from the straight line $3x - 5y + 7 = 0$ is

$$\left| \frac{3 \times 4 - 5 \times 5 + 7}{\sqrt{3^2 + 5^2}} \right| = \left| \frac{12 - 25 + 7}{\sqrt{34}} \right| = \frac{6}{\sqrt{34}}.$$

64. Let ABC be an equilateral triangle, where A is $(2, -1)$ and BC is $x + y = 2$.

Let AD be the length of perpendicular from A on BC .

$$\text{Thus, } AD = \frac{|2 - 1 - 2|}{\sqrt{1^2 + 1^2}} = \frac{1}{\sqrt{2}}$$

Therefore,

$$\sin(60^\circ) = \frac{AD}{AB}$$

$$\Rightarrow AB = \frac{2}{\sqrt{3}} \times \frac{1}{\sqrt{2}} = \sqrt{\frac{2}{3}}$$

Thus, the length of the side = $\sqrt{\frac{2}{3}}$

65. Equation of a straight line, whose intercepts are a and b is $\frac{x}{a} + \frac{y}{b} = 1$.

Given, the length of the perpendicular from the origin $= p$.

$$\Rightarrow \frac{|0+0-1|}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} = p$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2}$$

66. Clearly, both the lines are parallel
Hence, the required distance

$$= \frac{\left| \frac{11}{2} + 5 \right|}{\sqrt{3^2 + 4^2}} = \frac{|11+10|}{2.5} = \frac{21}{10} = 2\frac{1}{10} \text{ units.}$$

67. Equation of the line L is $\frac{x}{a} + \frac{y}{b} = 1$.

After rotation, the line L becomes, $\frac{x}{p} + \frac{y}{q} = 1$.

Therefore, the lengths of perpendicular from the origin to both the lines are same. Thus,

$$\frac{|0+0-1|}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} = \frac{|0+0-1|}{\sqrt{\frac{1}{p^2} + \frac{1}{q^2}}}$$

$$\Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2} + \frac{1}{q^2}$$

Hence, the result.

68. As we know that, area of a parallelogram whose sides are $a_1x + b_1y + d_1 = 0$, $a_2x + b_2y + c_2 = 0$ and $a_2x + b_2y + d_2 = 0$ is

$$= \frac{|(c_1 - d_1)(c_2 - d_2)|}{|a_1b_2 - a_2b_1|}$$

$$= \frac{|(4a + a) \times (7a - 3a)|}{|(1 \times (-2) - 3 \times 3)|} = \frac{20a^2}{11}$$

$$\text{Thus, } m = \frac{20}{11}.$$

$$\text{Now } 11m + 30 = 20 + 30 = 50$$

69. Given four lines are $ax + by + c = 0$,
 $ax + by - c = 0$, $ax - by + c = 0$
and $ax - by - c = 0$.

$$\text{Hence, the area} = \frac{|(c+c)(-c-c)|}{|(a \cdot (-b) - b \cdot a)|}$$

$$= \frac{|4c^2|}{|2ab|} = \frac{|2c^2|}{|ab|} = \frac{2c^2}{|ab|}$$

70. Hence, the points of intersection of the given lines is $(2, 6)$.

71. As we know that, the three lines $a_ix + b_iy + c_i = 0$, $i = 1, 2, 3$ are concurrent, if

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

$$= \begin{vmatrix} 2 & -3 & 5 \\ 3 & 4 & -7 \\ 9 & -5 & 8 \end{vmatrix}$$

$$= 2(32 - 35) + 3(24 + 63) + 5(-15 - 36)$$

$$= -6 + 261 - 255$$

$$= 0$$

Hence, the three lines are concurrent.

72. Equation of any line passing through the point of intersection of the lines

$$x + 3y - 8 = 0 \text{ and } 2x + 3y + 5 = 0 \text{ is}$$

$$(x - 3y + 8) + \lambda(2x + 3y + 5) = 0 \quad \dots(i)$$

which is passing through $(1, 2)$.

$$\text{Therefore, } (1 - 6 + 8) + \lambda(2 + 6 + 5) = 0$$

$$\Rightarrow \lambda = -\frac{3}{13}$$

From Eq. (i), we get

$$(x - 3y + 8) - \frac{3}{13}(2x + 3y + 5) = 0$$

$$\Rightarrow 13(x - 3y + 8) - 3(2x + 3y + 5) = 0$$

$$\Rightarrow 7x - 48y + 89 = 0$$

73. On solving $y = x + 1$ and $2x + y = 16$, we get,
 $x = 5$ and $y = 6$.

Thus, the point of concurrency is $(5, 6)$

Since, the given lines are concurrent, so

$$6 = 5m - 4$$

$$\Rightarrow 5m = 10$$

$$\Rightarrow m = 2$$

Hence, the value of m is 2.

74. Since the given lines $ax + y + 1 = 0$, $x + by + 1 = 0$ and $x + y + c = 0$ are concurrent,

$$\text{so } \begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & c \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} a-1 & 0 & 1 \\ 0 & b-1 & 1 \\ 1-c & 1-c & c \end{vmatrix} = 0$$

$$\Rightarrow (a-1)(c(b-1) - (1-c)) - (1-c)(b-1) = 0$$

$$\Rightarrow c(a-1)(b-1) - (a-1)(1-c) - (1-c)(b-1) = 0$$

$$\Rightarrow c(1-a)(1-b) + (1-a)(1-c) + (1-c)(1-b) = 0$$

$$\Rightarrow \frac{c}{1-c} + \frac{1}{1-b} + \frac{1}{1-a} = 0$$

$$\Rightarrow 1 + \frac{c}{1-c} + \frac{1}{1-b} + \frac{1}{1-a} = 1$$

$$\Rightarrow \frac{1}{1-a} + \frac{1}{1-b} + \frac{1}{1-c} = 1$$

75. Given family of lines are
 $ax + by + c = 0$... (i)

and also the given condition is
 $2a + 3b + 6c = 0$... (ii)

From Eqs (i) and (ii), we get

$$ax + by - \frac{a}{3} - \frac{b}{2} = 0$$

$$\Rightarrow a\left(x - \frac{1}{3}\right) + b\left(y - \frac{1}{2}\right) = 0$$

$$\Rightarrow \left(x - \frac{1}{3}\right) + \frac{b}{a}\left(y - \frac{1}{2}\right) = 0$$

$$\Rightarrow \left(x - \frac{1}{3}\right) = 0 \text{ and } \left(y - \frac{1}{2}\right) = 0$$

$$\Rightarrow x = \frac{1}{3} \text{ and } y = \frac{1}{2}$$

78. We have $4a^2 + 9b^2 - c^2 + 12ab = 0$

$$\Rightarrow (4a^2 + 9b^2 + 12ab) - c^2 = 0$$

$$\Rightarrow (2a^2 + 3b^2) - c^2 = 0$$

$$\Rightarrow (2a + 3b + c)(2a + 3b - c) = 0$$

$$\Rightarrow (2a + 3b + c) = 0 \text{ and } (2a + 3b - c) = 0$$

Given family of lines are

$$ax + by + c = 0 \quad \dots (i)$$

and also the given conditions are

$$(2a + 3b + c) = 0 \quad \dots (ii)$$

$$\text{and } (2a + 3b - c) = 0 \quad \dots (iii)$$

From Eqs (i) and (ii), we get

$$ax + by - 2a - 3b = 0$$

$$\Rightarrow a(x - 2) + b(y - 3) = 0$$

$$\Rightarrow (x - 2) + \frac{b}{a}(y - 3) = 0$$

$$\Rightarrow x = 2, y = 3$$

$$\Rightarrow m = 2, n = 3$$

From Eqs (i) and (iii), we get

$$ax + by + 2a + 3b = 0$$

$$\Rightarrow a(x + 2) + b(y + 3) = 0$$

$$\Rightarrow (x + 2) + \frac{b}{a}(y + 3) = 0$$

$$\Rightarrow x = -2, y = -3$$

$$\Rightarrow p = -2, q = -3$$

Thus, the value of

$$m + n + p + q + 10 = 2 + 3 - 2 - 3 + 10 = 10$$

81. Let m be the slope of the given line and the slope of $3x - y + 5 = 0$ is 3.

Therefore,

$$\tan 45^\circ = \left| \frac{m - 3}{1 + 3m} \right|$$

$$\Rightarrow \frac{m - 3}{1 + 3m} = \pm 1$$

$$\Rightarrow \frac{m - 3}{1 + 3m} = 1 \text{ and } \frac{m - 3}{1 + 3m} = -1$$

$$\Rightarrow 2m = -4 \text{ and } 4m = 2$$

$$\Rightarrow m = -2 \text{ and } m = \frac{1}{2}$$

82. Therefore,

$$\tan(45^\circ) = \left| \frac{m - \frac{1}{2}}{1 + \frac{m}{2}} \right|$$

$$\Rightarrow \left| \frac{2m - 1}{2 + m} \right| = 1$$

$$\Rightarrow \frac{2m - 1}{2 + m} = 1 \text{ and } \frac{2m - 1}{2 + m} = -1$$

$$\Rightarrow m = 3 \text{ and } m = -\frac{1}{3}$$

Hence, the equation of the lines are

$$y - 2 = 3(x - 3) \text{ and } y - 2 = -\frac{1}{3}(x - 3)$$

$$\Rightarrow 3x - y - 7 = 0 \text{ and } x + 3y - 9 = 0$$

83. Let ABC be an equilateral triangle, where A is $(2, 3)$ and BC is $x + y = 2$.

Equation of any line passing through $(2, 3)$ is

$$y - 3 = m(x - 2).$$

The slope of the line $x + y = 2$ is (-1) .

Therefore,

$$\tan(60^\circ) = \left| \frac{m + 1}{1 - m} \right|$$

$$\Rightarrow \frac{m + 1}{1 - m} = \pm \sqrt{3}$$

$$\Rightarrow \frac{m + 1}{1 - m} = \sqrt{3} \text{ and } \frac{m + 1}{1 - m} = -\sqrt{3}$$

$$\Rightarrow m(\sqrt{3} + 1) = (\sqrt{3} - 1) \text{ and } m(\sqrt{3} + 1) = (\sqrt{3} - 1)$$

$$\Rightarrow m = \left(\frac{\sqrt{3} - 1}{\sqrt{3} + 1} \right) \text{ and } m = \left(\frac{\sqrt{3} + 1}{\sqrt{3} - 1} \right)$$

$$\Rightarrow m = 2 - \sqrt{3} \text{ and } m = 2 + \sqrt{3}$$

Hence, the equations of the line are

$$y - 3 = (2 - \sqrt{3})(x - 2)$$

$$\text{and } y - 3 = (2 + \sqrt{3})(x - 2)$$

$$\Rightarrow (2 + \sqrt{3})x - y - (2\sqrt{3} + 1) = 0 \text{ and}$$

$$(2 - \sqrt{3})x - y + (2\sqrt{3} - 1) = 0$$

84. Let $AB: 4x + y = 1$,

$$AC: 3x - 4y + 1 = 0$$

and slope of AC is m .

The slope of $AB = -4$ and the slope of $AC = 3/4$.

Let $\angle ABC = \theta = \angle ACB$

$$\text{Thus, } \left(\frac{\frac{3-m}{4}}{1+\frac{3}{4}m} \right) = \left(\frac{-4-\frac{3}{4}}{1-4\cdot\frac{3}{4}} \right)$$

$$\Rightarrow \left(\frac{3-4m}{4+3m} \right) = \left(\frac{-16-3}{4-12} \right)$$

$$\Rightarrow \left(\frac{3-4m}{4+3m} \right) = \frac{19}{8}$$

$$\Rightarrow 57m + 76 = 24 - 32m$$

$$\Rightarrow 89m = 24 - 76 = -52$$

$$\Rightarrow m = -\frac{52}{89}$$

Hence, the equation of AC is

$$y + 7 = -\frac{52}{89}(x - 2)$$

$$\Rightarrow 52x + 89y + 519 = 0$$

85. Let PQ line intersects the line $4x + 3y = 12$ at A and $4x + 3y = 3$ at B, respectively.

Equation of any line PQ passing through $(-2, -7)$ is

$$y + 7 = m(x + 2) \quad \dots(i)$$

$$\text{Thus, } A = \left(\frac{33-6m}{4+3m}, \frac{20m-28}{4+3m} \right)$$

$$\text{and } B = \left(\frac{24-6m}{4+3m}, \frac{11m-28}{4+3m} \right)$$

$$\text{Given, } AB = 3 \Rightarrow AB^2 = 9$$

$$\Rightarrow \frac{81}{(4+3m)^2} + \frac{81m^2}{(4+3m)^2} = 9$$

$$\Rightarrow 9 + 9m^2 = 16 + 24m + 9m^2$$

$$\Rightarrow 24m = -7$$

$$\Rightarrow m = -7/24 \text{ and } m = \infty$$

Hence, the line PQ can be $x + 2 = 0$ and

$$\Rightarrow x + 2 = 0 \text{ and } 7x + 24y + 182 = 0$$

86. Equation of any line passing through $(2, 3)$ is

$$y - 3 = m(x - 2) \quad \dots(i)$$

Equation (i) is equally inclined with the lines

$$3x - 4y = 7 \quad \dots(ii)$$

$$\text{and } 12x - 5y + 6 = 0 \quad \dots(iii)$$

$$\text{Therefore, } \left(\frac{\frac{m-\frac{3}{4}}{1+\frac{3}{4}m}}{\frac{m-\frac{12}{5}}{1+\frac{12}{5}m}} \right) = -1$$

$$\Rightarrow \left(\frac{\frac{m-\frac{3}{4}}{1+\frac{3}{4}m}}{\frac{m-\frac{12}{5}}{1+\frac{12}{5}m}} \right) = -1$$

$$\Rightarrow \left(\frac{4m-3}{4+3m} \right) = \left(\frac{12-5m}{5+12m} \right)$$

$$\Rightarrow 63m^2 - 32m - 63 = 0$$

$$\Rightarrow 63m^2 - 81m + 49m - 63 = 0$$

$$\Rightarrow 9m(7m-9) + 7(7m-9) = 0$$

$$\Rightarrow (7m-9)(9m+7) = 0$$

$$\Rightarrow m = \frac{9}{7}, -\frac{7}{9}$$

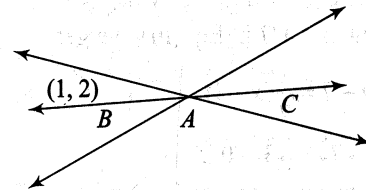
From Eqs (i), we get,

$$y - 3 = \frac{9}{7}(x - 2) \text{ and } y - 3 = -\frac{7}{9}(x - 2)$$

$$\Rightarrow 9x - 7y + 3 = 0 \text{ and } 7x + 9y = 41$$

which is the equations of the lines.

87.



$$\text{Let } AB: 3x + 4y = 5 \quad \dots(i)$$

$$\text{and } 4x - 3y = 15 \quad \dots(ii)$$

On solving Eqs (i) and (ii), we get

$$x = 3 \text{ and } y = 1$$

Thus, the co-ordinates of the point A be $(3, 1)$.

Equation of any line BC which passes through the point

$$(1, 2) \text{ is } y - 2 = m(x - 1) \quad \dots(iii)$$

On solving Eqs (i) and (iii), we get the co-ordinate B, i.e.

$$B = \left(\frac{4m-3}{4m+3}, \frac{2m+6}{4m+3} \right)$$

On solving Eqs (ii) and (iii), we get the co-ordinate C, i.e.

$$C = \left(\frac{21-3m}{4-3m}, \frac{11m+8}{4-3m} \right)$$

Given condition is $AB = AC$

$$\Rightarrow AB^2 = AC^2$$

$$\Rightarrow \left(\frac{21-3m}{4-3m} - 3 \right)^2 + \left(\frac{8+11m}{4-3m} + 1 \right)^2$$

$$= \left(\frac{4m-3}{4m+3} - 3 \right)^2 + \left(\frac{6+2m}{4m+3} + 1 \right)^2$$

$$\Rightarrow \frac{(9+6m)^2 + (12+8m)^2}{(4-3m)^2}$$

$$= \frac{(-8m-12)^2 + (6m+9)^2}{(4m+3)^2}$$

$$\Rightarrow (4m+3)^2(100m^2 + 300m + 225)$$

$$= (4-3m)^2(100m^2 + 300m + 225)$$

$$\Rightarrow (100m^2 + 300m + 225)$$

$$[(4m+3)^2 - (4-3m)^2] = 0$$

$$\Rightarrow (4m^2 + 12m + 9)[(4m+3)^2 - (4-3m)^2] = 0$$

$$\Rightarrow (2m+3)^2(m+7)(7m-1) = 0$$

$$\Rightarrow m = -\frac{3}{2}, -7, \frac{1}{7}$$

Put $m = -\frac{3}{2}$ in Eq. (iii), we get

$$y - 2 = -\frac{3}{2}(x - 1)$$

$$\Rightarrow 3x + 2y - 7 = 0$$

Clearly, it passes through $A(3, -1)$.

Thus, it can not be the equation of BC .

Put $m = -7$ in Eq. (iii), we get,

$$y = -7(x - 1) = -7x + 7$$

Thus, the equation of BC is $7x + y = 7$.

Also, put $m = 1/7$ in Eq. (iii), we get

$$y - 2 = \frac{1}{7}(x - 1)$$

$$\Rightarrow x - 7y + 13 = 0$$

Hence, the equations of BC can be $x - 7y + 13 = 0$ and $7x + y = 7$.

88. $3x + y + 7 = 0$ and $x - 3y = 31$.

Do yourself.

89. Hence, the equations of the bisectors are

$$\left| \frac{3x - 4y + 7}{\sqrt{3^2 + 4^2}} \right| = \left| \frac{12x + 5y - 2}{\sqrt{12^2 + 5^2}} \right|$$

$$\Rightarrow \left(\frac{3x - 4y + 7}{5} \right) = \pm \left(\frac{12x + 5y - 2}{13} \right)$$

$$\Rightarrow 13(3x - 4y + 7) = 5(12x + 5y - 2)$$

$$\Rightarrow 13(3x - 4y + 7) = 5(12x + 5y - 2)$$

$$\text{and } 13(3x - 4y + 7) = -5(12x + 5y - 2)$$

$$\Rightarrow 21x + 77y = 101 \text{ and } 3x - y + 3 = 0$$

90. Hence, the equation of the bisector, containing the origin is

$$\left(\frac{-4x - 3y + 6}{\sqrt{4^2 + 3^2}} \right) = \left(\frac{5x + 12y + 9}{\sqrt{5^2 + 12^2}} \right)$$

$$\Rightarrow \left(\frac{-4x - 3y + 6}{5} \right) = \left(\frac{5x + 12y + 9}{13} \right)$$

$$\Rightarrow 13(-4x - 3y + 6) = 5(5x + 12y + 9)$$

$$\Rightarrow 7x + 9y = 3$$

91. The given lines are $-2x - y + 6 = 0$

and $2x - 4y + 7 = 0$.

Therefore, $-2 \cdot 1 - 2 + 6 = -4 + 6 = 2$

and $2 \cdot 1 - 4 \cdot 2 + 7 = 9 - 8 = 1$

Thus, positive one is the equation of the bisector.

Hence, the equation of bisector is

$$\left(\frac{-2x - y + 6}{\sqrt{2^2 + 1^2}} \right) = \left(\frac{2x - 4y + 7}{\sqrt{2^2 + 4^2}} \right)$$

$$\Rightarrow \left(\frac{-2x - y + 6}{\sqrt{5}} \right) = \left(\frac{2x - 4y + 7}{2\sqrt{5}} \right)$$

$$\Rightarrow 2(-2x - y + 6) = (2x - 4y + 7)$$

$$\Rightarrow 6x - 2y = 5$$

92. The given lines are $3x - 4y + 7 = 0$

and $-12x - 5y + 2 = 0$.

Now, $a_1a_2 + b_1b_2 = -36 + 20 = -16 < 0$.

Therefore, negative one is the obtuse-angle bisector.

Hence, the equation of the bisector of the obtuse angle is

$$\left(\frac{3x - 4y + 7}{\sqrt{3^2 + 4^2}} \right) = - \left(\frac{-12x - 5y + 2}{\sqrt{12^2 + 5^2}} \right)$$

$$\Rightarrow \left(\frac{3x - 4y + 7}{5} \right) = - \left(\frac{-12x - 5y + 2}{13} \right)$$

$$\Rightarrow 21x + 77y = 101$$

93. The given lines are $-x - y + 3 = 0$ and $7x - y + 5 = 0$.

Now, $a_1a_2 + b_1b_2 = -7 + 1 = -6 < 0$.

Therefore, positive one is the acute-angle bisector.

Hence, the equation of the acute-angle bisector is

$$\left(\frac{-x - y + 3}{\sqrt{1^2 + 1^2}} \right) = \left(\frac{7x - y + 5}{\sqrt{7^2 + 1^2}} \right)$$

$$\Rightarrow \left(\frac{-x - y + 3}{\sqrt{2}} \right) = \left(\frac{7x - y + 5}{5\sqrt{2}} \right)$$

$$\Rightarrow 5(-x - y + 3) = (7x - y + 5)$$

$$\Rightarrow 6x - 2y = 5$$

94. The lengths of perpendicular from any point on the line

$7x - 9y + 10 = 0$ to the lines $3x + 4y = 5$ and $12x + 5y = 7$ are same if $7x - 9y + 10 = 0$ is the bisector of

$3x + 4y = 5$ and $12x + 5y = 7$.

Hence, the equation of bisectors is

$$\left(\frac{-3x - 4y + 5}{\sqrt{3^2 + 4^2}} \right) = \left(\frac{-12x - 5y + 7}{\sqrt{12^2 + 5^2}} \right)$$

$$\Rightarrow 13(-3x - 4y + 5) = 5(-12x - 5y + 7)$$

$$\Rightarrow 21x - 27y + 30 = 5$$

$$\Rightarrow 7x - 9y + 10 = 0$$

Hence, the result.

95. Let ABC be a triangle,

where $AB: x + 1 = 0$, $BC: 3x - 4y = 5$ and $CA: 5x + 12y = 27$.

Case I: Acute-angle bisector between AB and AC

The given lines are $x + 1 = 0$ and $-3x + 4y + 5 = 0$.

Now, $a_1a_2 + b_1b_2 = 1 \cdot (-3) + 0 = -3 < 0$.

Thus, positive one is the acute-angle bisector.

Hence, the acute-angle bisector is

$$\left(\frac{x + 1}{\sqrt{1^2}} \right) = \left(\frac{-3x + 4y + 5}{\sqrt{3^2 + 4^2}} \right)$$

$$\Rightarrow \left(\frac{x + 1}{1} \right) = \left(\frac{-3x + 4y + 5}{5} \right)$$

$$\Rightarrow 5(x + 1) = (-3x + 4y + 5)$$

$$\Rightarrow 2x - y = 5$$

...(i)

Case II: The acute-angle bisector between BC and AC .

The given lines are $-3x + 4y + 5 = 0$

and $-5x - 12y + 27 = 0$.

Now, $a_1a_2 + b_1b_2 = 15 - 48 = -23 < 0$

Thus, the positive one is the acute-angle bisector.

Hence, the acute-angle bisector is

$$\begin{aligned} \left(\frac{-3x + 4y + 5}{\sqrt{3^2 + 4^2}} \right) &= \left(\frac{-5x - 12y + 27}{\sqrt{5^2 + 12^2}} \right) \\ \Rightarrow \left(\frac{-3x + 4y + 5}{5} \right) &= \left(\frac{-5x - 12y + 27}{13} \right) \\ \Rightarrow x - 8y &= -5 \quad \dots(ii) \end{aligned}$$

On solving Eqs (i) and (ii), we get

$$x = \frac{1}{3}, y = \frac{2}{3}.$$

Hence, the in-centre is $\left(\frac{1}{3}, \frac{2}{3} \right)$.

96. The given lines are $\sqrt{3}x - y + 3 = 0$ and $x - \sqrt{3}y + 3\sqrt{3} = 0$

Hence, the equation of bisectors are

$$\left| \frac{\sqrt{3}x - y + 3}{\sqrt{3+1}} \right| = \left| \frac{x - \sqrt{3}y + 3\sqrt{3}}{\sqrt{1+3}} \right|$$

and $(\sqrt{3}x - y + 3) = -(x - \sqrt{3}y + 3\sqrt{3})$
 $(\sqrt{3} - 1)x + (\sqrt{3} - 1)y = 3(\sqrt{3} - 1)$

$$\Rightarrow \text{and } (\sqrt{3} + 1)x + (\sqrt{3} + 1)y = -3(\sqrt{3} + 1)$$

$$\Rightarrow x + y = 3 \text{ and } x + y = -3$$

Thus, the point P is $(3, 0)$ and the point Q is $(-3, 0)$

Hence, the length of $PQ = 6$.

97. Give lines are $AB: x + y = 1$...(i)
 and $CD: x + y = 5$...(ii)

Slope of $AB = -1 = \tan(135^\circ)$

Since, AB makes an angle of 45° with AC , therefore it will make an angle of $(135^\circ - 45^\circ)$ or $(135^\circ + 45^\circ)$, i.e. 90° or 180° with the positive direction of x -axis.

Thus, AB is parallel to x -axis or y -axis.

Hence, $x - 2 = 0$ and $y + 1 = 0$.

Case I: Consider the lines are $-x - y + 1 = 0$

and $-x + 2 = 0$

Now, $a_1a_2 + b_1b_2 = 1 + 0 = 1 > 0$

Thus negative one is the acute-angle bisector.

Hence, the equation of the acute-angle bisector of $\angle BAC$ is

$$\begin{aligned} \left(\frac{-x - y + 1}{\sqrt{1^2 + 1^2}} \right) &= - \left(\frac{-x + 2}{\sqrt{1^2}} \right) \\ \Rightarrow (\sqrt{2} + 1)x + y &= (2\sqrt{2} + 1) \quad \dots(iii) \end{aligned}$$

Case II: Consider the lines are $-x - y + 1 = 0$ and $y + 1 = 0$

Now, $a_1a_2 + b_1b_2 = 0 - 1 = -1 < 0$

Thus, positive one is the acute-angle bisector.

Hence, the equation of the acute-angle bisector of $\angle BAC$ is

$$\begin{aligned} \left(\frac{-x - y + 1}{\sqrt{1^2 + 1^2}} \right) &= \left(\frac{y + 1}{\sqrt{1^2}} \right) \\ \Rightarrow x + (\sqrt{2} + 1)y &= (1 - \sqrt{2}) \quad \dots(iv) \end{aligned}$$

On solving Eqs (i) and (iii), we get

$$x = 6 + 2\sqrt{2}, y = -1 - 2\sqrt{2}$$

Thus, the co-ordinates of C can be

$$(6 + 2\sqrt{2}, 1 - 2\sqrt{2})$$

Again, solving Eqs (i) and (iv), we get

$$x = 2 - 2\sqrt{2}, y = 3 + 2\sqrt{2}$$

Thus, The co-ordinates of C may be

$$(2 - 2\sqrt{2}, 3 + 2\sqrt{2})$$

98. $(-1/10, 37/10)$

99. Let the image of the point $(-8, 12)$ be (x_2, y_2) .

$$\begin{aligned} \text{Then } \frac{x_2 + 8}{4} &= \frac{y_2 - 12}{7} = - \frac{2(4 \cdot -8 + 7 \cdot 12 + 13)}{(4^2 + 7^2)} \\ &= \frac{2(-32 + 84 + 13)}{65} \\ &= \frac{2 \times 65}{65} = 2 \end{aligned}$$

$$\Rightarrow \frac{x_2 + 8}{4} = 2 \text{ and } \frac{y_2 - 12}{7} = 2$$

$$\Rightarrow x_2 = 0 \text{ and } y_2 = 28$$

Hence, the image of the point $(-8, 12)$ is $(0, 28)$.

100. Hence, the image of the point $(3, 4)$ with respect to the line $y = x$ is $(4, 3)$.

101. Let the point $(4, 2)$ be P and the point $(-2, 6)$ be Q .

Now the mid-point of P and Q is $M(1, 4)$.

Equation of a line passing through $(4, 2)$ and $(-2, 6)$ is

$$y - 2 = \frac{6 - 2}{-2 - 4}(x - 4) = -\frac{2}{3}(x - 4)$$

$$\Rightarrow 3y - 6 = -2x + 4$$

$$\Rightarrow 2x + 3y = 10 \quad \dots(i)$$

Equation of any line perpendicular to Eq. (i) is

$$3x - 2y + k = 0$$

which is passing through $M(1, 4)$.

So $k = 8 - 3 = 5$.

Hence, the equation of the line is

$$3x - 2y + 5 = 0.$$

102. Let the point M be $(4, 1)$.

The image of the point $M(1, 4)$ with respect to the line $y = x$ be $P(4, 1)$.

Now, if the point P be translated about the line $x = 2$, then the new position of P is $(4 + 2, 1)$.

Hence, the co-ordinates of Q is $(6, 1)$.

103. Let the point P be $(3, 2)$.

The image of the point P with respect to the line $x = 4$ is $Q(2 \cdot 4 - 3, 2)$ i.e. $Q(5, 2)$.

Let $Z = 5 + 2i$ and R be Z_1

Now, by rotation theorem of complex number,

$$\Rightarrow \frac{Z_1 - 0}{Z - 0} = \frac{|Z - 0|}{|Z_1 - 0|} e^{i\frac{\pi}{4}}$$

$$\Rightarrow \frac{Z_1}{Z} = \frac{|Z_1 - 0|}{|Z - 0|} e^{i\frac{\pi}{4}} = e^{i\frac{\pi}{4}}$$

$$\Rightarrow Z_1 = Z \times e^{i\frac{\pi}{4}} = (5+2i) \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right)$$

$$\Rightarrow Z_1 = \frac{1}{\sqrt{2}} (5+5i+2i-2) = \frac{1}{\sqrt{2}} (3+7i)$$

Hence, the co-ordinates of R are $\left(\frac{3}{\sqrt{2}}, \frac{7}{\sqrt{2}} \right)$.

104.

105. Let $PM: x - 2y + 3 = 0$ be the incident ray and MR be the reflected ray. Produced MQ in such a way that $PM = MQ$. Now we shall find the image of Q w.r.t. AB is R . Co-ordinates of M are $(1, -1)$.

Let co-ordinates of Q be $(3, 0)$ and R be (h, k) .

Thus, R is the image of Q with respect to $3x - 2y = 5$.

$$\text{Now } \frac{h-3}{3} = \frac{k-0}{-2} = -\frac{2(3.3+2.0-5)}{(3^2+2^2)} = -\frac{8}{13}$$

$$\Rightarrow h = \frac{15}{13} \quad \text{and} \quad k = \frac{16}{13}$$

Thus, the co-ordinates of R be $\left(\frac{15}{13}, \frac{16}{13} \right)$.

$$\text{Slope of } MR = \frac{29}{2}$$

Therefore, the equation of MR is

$$y+1 = \frac{29}{2}(x-1)$$

$$\Rightarrow 29x - 2y = 31$$

Hence, the equation of the reflected ray is

$$29x - 2y = 31$$

106. Let N be the image of M with respect to x -axis. Thus N is $(5, -3)$

$$\text{Slope of } NQ = \frac{-3-2}{5-1} = -\frac{5}{4}$$

Therefore, the equation QN is

$$y-2 = -\frac{5}{4}(x-1)$$

$$\Rightarrow 5x + 4y = 13$$

Thus, the co-ordinates of P be $\left(\frac{13}{5}, 0 \right)$.

Hence, the abscissa of the point P is $\frac{13}{5}$.

107. Let PM be the incident ray and QM be the reflected ray.

Let $P(1, 5)$ be any point on the line $x = 1$

Produce PM in such a way that $PM = RM$,

where $M = (1, 0)$

Clearly, $R = (1, -5)$, since, the incident ray is parallel to y -axis.

Now, image of R with respect to the line $x + y = 1$ is Q .

Let the co-ordinate of Q be (h, k) .

$$\text{Now, } \frac{h-1}{1} = \frac{k+5}{1} = \frac{-2(1-5-1)}{(1+1)} = 5$$

$$\Rightarrow h = 6, k = 0$$

Thus, the co-ordinates of Q is $(6, 0)$.

Since, the co-ordinates of M is $(1, 0)$ and the co-ordinates of Q is $(6, 0)$, so the equation of the reflected ray is $y = 0$.

108. Let $PM: x - 6y = 8$ be the incident ray

$AB: x + y = 1$ is the mirror, MQ is the reflected ray and MS is the refracted ray.

Clearly, M is $(2, -1)$

$$\text{Now, } \tan 15^\circ = \left| \frac{m - \frac{1}{6}}{1 + \frac{m}{6}} \right|$$

$$\Rightarrow (2 - \sqrt{3}) = \pm \left(\frac{6m - 1}{m + 6} \right)$$

$$\Rightarrow \left(\frac{6m - 1}{m + 6} \right) = 2 - \sqrt{3} \quad \text{or} \quad \sqrt{3} - 2$$

$$\Rightarrow m = \frac{70 - 37\sqrt{3}}{13} \quad \text{or} \quad \frac{37\sqrt{3} - 70}{13}$$

Let the angle between $x + y = 1$ and the line through $M(2, -1)$ with the slope

$$m = \frac{70 - 37\sqrt{3}}{13} \quad \text{be } \alpha.$$

$$\text{Then } \tan(\alpha) = \left| \frac{\frac{70 - 37\sqrt{3}}{13} + 1}{1 - \frac{70 - 37\sqrt{3}}{13}} \right| = \left| \frac{83 - 37\sqrt{3}}{37\sqrt{3} - 57} \right|$$

$$= \frac{83 - 37\sqrt{3}}{37\sqrt{3} - 57}$$

and the angle between $x + y = 1$ and the line through

$M(2, -1)$ with slope $m = \frac{37\sqrt{3} - 70}{13}$ be β .

$$\text{Then } \tan(\beta) = \left| \frac{\frac{37\sqrt{3} - 70}{13} + 1}{1 - \frac{37\sqrt{3} - 70}{13}} \right| = \left| \frac{37\sqrt{3} - 9}{131 - 37\sqrt{3}} \right|$$

$$= \frac{37\sqrt{3} - 9}{131 - 37\sqrt{3}}$$

Here, $\tan(\alpha) > \tan(\beta)$

$$\Rightarrow \alpha > \beta$$

Therefore, the slope of the refracted ray is $\frac{70 - 37\sqrt{3}}{13}$.

Hence, the equation of the refracted ray is

$$(y+1) = \left(\frac{70 - 37\sqrt{3}}{13} \right)(x-2)$$

$$\Rightarrow 13y + 13 = (70 - 37\sqrt{3})x - 140 + 74\sqrt{3}$$

$$\Rightarrow (70 - 37\sqrt{3})x - 13y - 153 + 74\sqrt{3} = 0$$